

All India Aakash Test Series for NEET - 2025

TEST - 7 (Code-E)**Click here for
Code-F sol.**

Test Date : 23/03/2025

ANSWERS

1. (4)	37. (3)	73. (2)	109. (4)	145. (1)
2. (2)	38. (3)	74. (2)	110. (4)	146. (3)
3. (3)	39. (1)	75. (2)	111. (2)	147. (4)
4. (1)	40. (4)	76. (2)	112. (4)	148. (3)
5. (4)	41. (4)	77. (3)	113. (1)	149. (3)
6. (4)	42. (1)	78. (3)	114. (3)	150. (2)
7. (3)	43. (2)	79. (3)	115. (4)	151. (4)
8. (3)	44. (4)	80. (2)	116. (2)	152. (1)
9. (4)	45. (3)	81. (1)	117. (3)	153. (4)
10. (4)	46. (4)	82. (4)	118. (3)	154. (1)
11. (3)	47. (1)	83. (4)	119. (2)	155. (2)
12. (2)	48. (2)	84. (4)	120. (3)	156. (3)
13. (1)	49. (1)	85. (2)	121. (3)	157. (2)
14. (4)	50. (2)	86. (2)	122. (2)	158. (4)
15. (4)	51. (4)	87. (3)	123. (1)	159. (2)
16. (2)	52. (3)	88. (2)	124. (3)	160. (2)
17. (1)	53. (1)	89. (1)	125. (1)	161. (2)
18. (4)	54. (1)	90. (3)	126. (4)	162. (2)
19. (1)	55. (3)	91. (2)	127. (4)	163. (4)
20. (3)	56. (4)	92. (4)	128. (1)	164. (2)
21. (2)	57. (4)	93. (3)	129. (3)	165. (3)
22. (4)	58. (2)	94. (3)	130. (4)	166. (1)
23. (1)	59. (4)	95. (4)	131. (2)	167. (2)
24. (3)	60. (2)	96. (1)	132. (4)	168. (1)
25. (4)	61. (3)	97. (4)	133. (3)	169. (3)
26. (3)	62. (4)	98. (3)	134. (3)	170. (1)
27. (1)	63. (4)	99. (3)	135. (3)	171. (2)
28. (1)	64. (2)	100. (4)	136. (2)	172. (3)
29. (2)	65. (1)	101. (1)	137. (2)	173. (3)
30. (3)	66. (2)	102. (1)	138. (3)	174. (1)
31. (1)	67. (3)	103. (4)	139. (3)	175. (4)
32. (2)	68. (2)	104. (4)	140. (3)	176. (2)
33. (4)	69. (4)	105. (1)	141. (1)	177. (1)
34. (2)	70. (4)	106. (3)	142. (2)	178. (2)
35. (2)	71. (4)	107. (3)	143. (3)	179. (2)
36. (3)	72. (1)	108. (3)	144. (4)	180. (1)

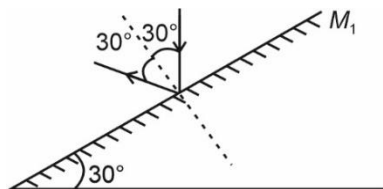
HINTS & SOLUTIONS

[PHYSICS]

1. Answer (4)

Hint: According to law of reflection, angle of incidence is equal to angle of reflection.

Sol.:



Angle of reflection = 30°

2. Answer (2)

Hint: At the instant of closest approach, α -particles is momentarily at rest.

$$\text{Sol.: } \therefore \frac{1}{4\pi\epsilon_0} \frac{(Ze)(2e)}{r_0} = \frac{1}{2}mv^2$$

$$\Rightarrow r_0 = \frac{1}{\pi\epsilon_0} \times \frac{Ze^2}{mv^2}, \Rightarrow r_0 \propto \frac{1}{m}$$

3. Answer (3)

Hint: If virtual and erect image is formed so m is +ve $\Rightarrow m = +3$

$$\text{Sol.: } \frac{f}{f-u} = 3$$

$$u = \frac{-20}{3} \text{ cm}$$

4. Answer (1)

Hint: For n-type, electrons are majority charge carriers.

Sol.: In an extrinsic semiconductor doped with pentavalent impurity, electrons become the majority carriers and holes the minority carriers.

$$n_e \gg n_h$$

5. Answer (4)

Hint & Sol.: Energy is released when BE/nucleon is increased.

6. Answer (4)

Hint & Sol.: In interference and diffraction, light energy is redistributed. If it reduces in one region, producing a dark fringe, it increases in another region, producing a bright fringe. There is no gain or loss of energy, which is consistent with the principle of conservation of energy.

7. Answer (3)

Hint & Sol.: When light travels from rarer to a denser medium, the speed decreases and frequency remain same.

8. Answer (3)

Hint: Power (P) transmitted in the beam = $NE = Nh\nu$

$$\text{Sol.: } P = 2 \times 10^{16} \times 6.63 \times 10^{-34} \times 5 \times 10^{14} \\ = 6.63 \times 10^{-3} \text{ W} = 6.63 \text{ mW}$$

9. Answer (4)

$$\text{Hint: } I = 4I_0 \cos^2\left(\frac{\Delta\phi}{2}\right)$$

$$\text{Sol.: If } \Delta x = \frac{\lambda}{3}, \Delta\phi = \frac{2\pi}{3}, I = 4I_0 \cos^2 60^\circ = I_0$$

$$\text{If } \Delta x = \frac{\lambda}{6}, \Delta\phi = \frac{\pi}{3}, I = 4I_0 \cos^2 30^\circ = 3I_0$$

$$\text{If } \Delta x = \frac{\lambda}{4}, \Delta\phi = \frac{\pi}{2}, I = 4I_0 \cos^2 \frac{\pi}{4} = 2I_0$$

$$\text{If } \Delta x = \frac{\lambda}{2}, \Delta\phi = \pi, I = 4I_0 \cos^2 \frac{\pi}{2} = \text{Zero}$$

10. Answer (4)

Hint: When an object is placed at the second focus of a concave lens, the image will be formed at infinity.

$$\text{Sol.: For first lens, } \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} - \frac{1}{-15} = \frac{1}{10}$$

$$\Rightarrow v = 30 \text{ cm}$$

Virtual object for second lens at 25 cm from this lens,

$$\frac{1}{v'} - \frac{1}{25} = \frac{-1}{25} \Rightarrow v' = \infty$$

11. Answer (3)

Hint & Sol.: For a given frequency of the incident radiation, the stopping potential is independent of its intensity. The maximum kinetic energy of photoelectrons depends on frequency of the light source and the emitter plate material, but is independent of intensity of incident radiation.

12. Answer (2)

Hint: Use Einstein's equation of photoelectric effect, $K_{\max} = h\nu - \phi$

$$\text{Sol.: } 1.2 = h\nu - \phi \quad (i)$$

$$3.6 = 1.5 h\nu - \phi \quad (ii)$$

On solving equation (i) and (ii)

$$\therefore \phi = 3.6 \text{ eV}$$

13. Answer (1)

Hint & Sol.: As temperature rises, the atoms of the liquid become more mobile and the coefficient of viscosity η falls. In a gas as temperature rises, it increases the random motion of atoms and as a consequence viscosity of gases increases.

14. Answer (4)

Hint & Sol.: In explosion $\Rightarrow$ Momentum remain conserved

$$\lambda = \frac{h}{p}$$

$$\text{Since } p_1 = p_2 \Rightarrow \lambda_1 = \lambda_2$$

15. Answer (4)

Hint & Sol.: Between two nucleons, the nuclear force is attractive for separation larger than 0.8 fm. The nuclear mass is always less than the total mass of its constituents.

16. Answer (2)

$$\text{Hint: P.E.} = \frac{-kZe^2}{r}$$

$$\text{T.E.} = \frac{\text{P.E.}}{2}$$

Sol.: As principal quantum number (n) increases, the radius of the orbit increases and potential energy becomes less negative (increases). As n increases, T.E. also increases

17. Answer (1)

$$\text{Hint: } \frac{1}{\lambda} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

Sol.: For shortest wavelength of Lyman series:

$$n_1 = 1, n_2 = \infty$$

$$\frac{1}{\lambda} = R$$

$$\lambda = \frac{1}{R}$$

18. Answer (4)

Hint: Properties of semiconductors.

Sol.: For insulators $E_g > 3 \text{ eV}$, for semiconductor E_g is 0.2 eV to 3 eV while for metals $E_g \approx 0$

19. Answer (1)

Hint: P.E. of hydrogen atom in ground state is -27.2 eV .

Sol.: If the P.E. in ground state is taken as zero then whole system is shifted by 27.2 eV. So, T.E. in first excited state will be

$$(-3.4 + 27.2) \text{ eV} \\ = 23.8 \text{ eV}$$

20. Answer (3)

Hint & Sol.: The minimum energy to ionise an atom is the energy required to remove the most loosely bound (outermost) electron.

21. Answer (2)

$$\text{Hint: } m = \frac{f_0}{f_e}$$

$$\text{Sol.: } m = \frac{100}{1} = 100$$

22. Answer (4)

Hint & Sol.: Properties of intrinsic semiconductor.

23. Answer (1)

Hint: Use formula $R = R_0 A^{1/3}$

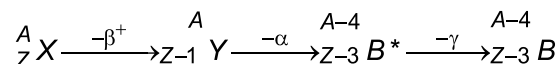
$$\text{Sol.: } R = R_0 A^{1/3} \\ = 1.2 \times (70)^{1/3} \\ = 4.94 \text{ fm}$$

24. Answer (3)

Hint: In β^+ decay, atomic number decreases by 1 and mass number remains same.

In decay process when product nucleus is in its excited state, then it emits a γ -photon (γ -decay)

Sol.:

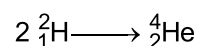


25. Answer (4)

Hint & Sol.: For any semiconductor: $n_i^2 = n_e n_h$

26. Answer (3)

Hint: Given reaction represented as



Sol.: Energy released (Q-value)

$$= (B.E.)_{\text{product}} - (B.E.)_{\text{reactant}} \\ = b - 2a$$

27. Answer (1)

Hint: An intrinsic semiconductor will behave like an insulator at $T = 0\text{ K}$ and at $T > 0\text{ K}$, some thermally generated electron hole pairs are to be depicted.

Sol.: It is the thermal energy at higher temperatures ($T > 0\text{ K}$), which excites some electrons from the valence band to conduction band. These thermally excited electrons at $T > 0\text{ K}$, partially occupy the conduction band. Therefore, the energy-band diagram should be option (1)

28. Answer (1)

Hint: Use Snell's law

$$n_1 \sin i = n_2 \sin r$$

$$\text{Sol.: } n_1 \sin i = n_2 \sin r_1$$

$$n_2 \sin r_1 = n_3 \sin r_2$$

$$n_3 \sin r_2 = n_4 \sin r_3$$

$$n_4 \sin r_3 = n_5 \sin i$$

Comparing all equations

$$n_1 \sin i = n_5 \sin i$$

If $n_1 = n_5$ then incident ray and emergent ray are parallel to each other.

29. Answer (2)

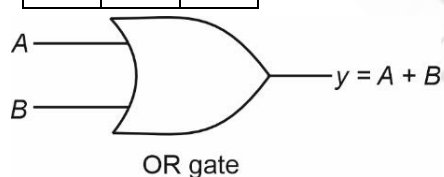
Hint & Sol.: Nuclides with same neutron number N but different atomic number Z are, isotones.

30. Answer (3)

Hint: When both A and B are open then we cannot get any output.

Sol.:

A	B	C
0	0	0
0	1	1
1	0	1
1	1	1



31. Answer (1)

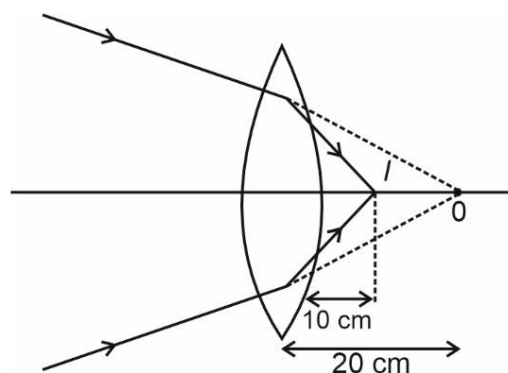
Hint: $E_n = \frac{-13.6Z^2}{n^2} \text{ eV}$

Sol.: $E_3 - E_1$ energy difference is greatest among the given options in the question.

$\therefore$ In case-I, most energetic photon will emit.

32. Answer (2)

Hint & Sol.:



By lens formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{+10} - \frac{1}{20} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{20} = \frac{1}{f}$$

$$\Rightarrow f = 20 \text{ cm}$$

33. Answer (4)

Hint & Sol.: Reflection, interference and diffraction can all be explained by wave phenomena.

34. Answer (2)

Hint & Sol.: Searle's apparatus is used for the measurement of Young's modulus.

35. Answer (2)

Hint: Fringe width $\beta = \frac{\lambda D}{d}$

Sol.: $\frac{\beta_2}{\beta_1} = \frac{\lambda_2}{\lambda_1} \Rightarrow \lambda_2 = 560 \text{ nm}$

36. Answer (3)

Hint: Use Einstein's photoelectric equation.

Sol.: $k_1 = h\nu - \phi$

$$k_2 = 2h\nu - \phi$$

$$k_2 = 2(k_1 + \phi) - \phi$$

$$= 2k_1 + \phi$$

$$\frac{1}{2} M v_2^2 = 2 \times \frac{1}{2} M v_1^2 + \phi$$

$$v_2 = \sqrt{2v_1^2 + \frac{2\phi}{M}}$$

37. Answer (3)

Hint: For a metre bridge at null point $\frac{P}{Q} = \frac{l}{100-l}$

Sol.: $\frac{P}{Q} = \frac{60}{40}$

$$\frac{6}{Q} = \frac{60}{40}$$

$$Q = 4 \Omega$$

38. Answer (3)

Hint: $\frac{P}{Q} = \frac{l}{100-l}$

Sol.: $\frac{6}{3} = \frac{2}{1} = \frac{l}{100-l}$

$$\text{Null point} = 66.7 \text{ cm}$$

$$\Rightarrow \text{Shift} = 6.7 \text{ cm}$$

39. Answer (1)

Hint: $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

Sol.: $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

$$\frac{3}{2v} + \frac{1}{15} = \frac{0.5}{-30}$$

$$\Rightarrow \frac{3}{2v} = \frac{1}{-60} - \frac{4}{60} = \frac{-5}{60}$$

$$v = -18 \text{ cm (virtual image)}$$

40. Answer (4)

Hint & Sol.: Large line source will produce cylindrical wavefront.

41. Answer (4)

Hint & Sol.: Energy required to remove an e^- from ground state of hydrogen atom is 13.6 eV

$$\therefore r = r_0 = 0.53 \text{ \AA}$$

42. Answer (1)

Hint: Use, $eV_0 = h\nu - \phi_0$

Sol.: $V_0 = \frac{h}{e} \nu - \frac{\phi_0}{e}$

$$\therefore \text{slope } m = \frac{h}{e}$$

43. Answer (2)

Hint: $2\pi r = n\lambda$ and $r_n = \frac{0.529n^2 \text{ \AA}}{Z}$

Sol.: $n = 2$ (first excited state)

$$Z = 1$$

$$\lambda = \pi r_2$$

$$\lambda = \pi(0.529 \text{ \AA}) \frac{(2^2)}{1}$$

$$\lambda = 2.12\pi \text{ \AA}$$

44. Answer (4)

Hint: $\sin C = \frac{1}{\mu} = \frac{1}{\sqrt{3}}$ where C is critical angle of face AC.

Sol.: $i = 60^\circ$ on face AC.

$$\sin i > \sin C \quad \left(\because \frac{\sqrt{3}}{2} > \frac{1}{\sqrt{3}} \right)$$

$$\Rightarrow i > C$$

Ray undergoes TIR on face AC.

Angle of emergence doesn't exist.

45. Answer (3)

Hint & Sol.: In forward biasing, p -side of diode is connected to positive terminal of battery and n -side is connected to negative terminal of battery.

In forward biasing, width of depletion layer decreases.

[CHEMISTRY]

46. Answer (4)

Hint: Lactose is a disaccharide which is found in milk.

Sol.: Lactose on hydrolysis gives β -D-galactose and β -D-glucose. It is a reducing sugar.

47. Answer (1)

Hint: Glucose does not form the hydrogen sulphite addition product with NaHSO_3

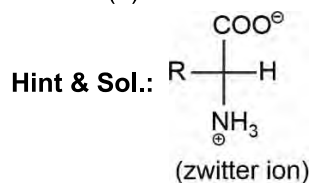
Sol.: Cellulose is a straight chain polysaccharide composed only of β -D-glucose units.

48. Answer (2)

Hint: The amino acids, which can be synthesised in the body, are known as non-essential amino acid.

Sol.: Tyrosine is a non-essential amino acid.

49. Answer (1)



50. Answer (2)

Hint & Sol.: During denaturation of proteins, secondary and tertiary structure are destroyed but primary structure remains intact.

51. Answer (4)

Hint: Complete hydrolysis of DNA yields a pentose sugar, phosphoric acid and nitrogen containing heterocyclic compound.

Sol.: Sugar moiety present in RNA molecule is β -D-ribose.

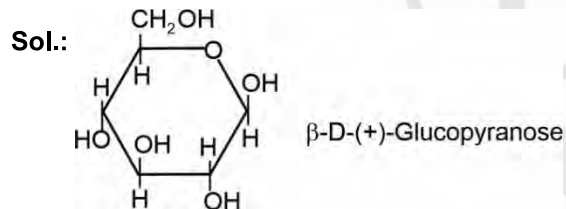
52. Answer (3)

Hint: Vitamin A, D, E and K are fat soluble.

Sol.: B group vitamins and vitamin C are water soluble.

53. Answer (1)

Hint: β -D-(+)-Glucopyranose contains five chiral carbon atoms.



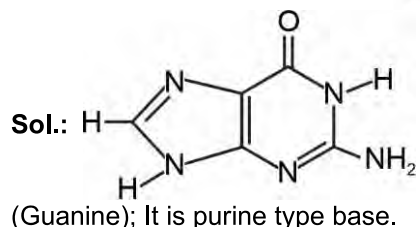
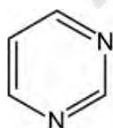
54. Answer (1)

Hint: In amylopectin, branching occurs by C1-C6 glycosidic linkage.

Sol.: Amylose is water soluble components which constitutes about 15-20% of starch. Amylopectin is insoluble in water and constitutes about 80-85% of starch. It is a branched chain polymer of α -D-glucose units in which chain is formed by C1-C4 glycosidic linkage whereas branching occurs by C1-C6 glycosidic linkage.

55. Answer (3)

Hint: Structure of pyrimidine is



56. Answer (4)

Hint: Compound having vacant orbital act as Lewis's acid.

Sol.: NH_3 act as Lewis base due to presence of lone pair.

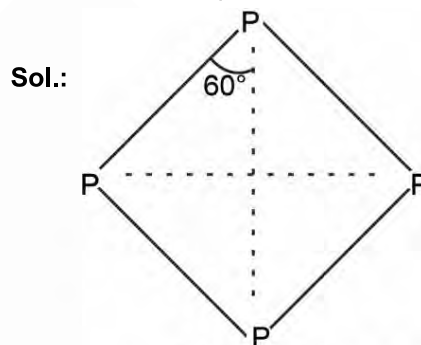
57. Answer (4)

Hint: Borax is $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$

Sol.: In borax two boron atoms are sp^2 hybridised while other two are sp^3 hybridised.

58. Answer (2)

Hint: Red phosphorus is insoluble in water as well as in carbon disulphide.



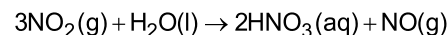
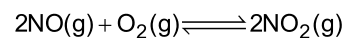
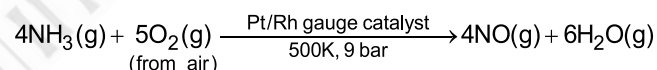
White phosphorus

Red phosphorus is obtained by heating white phosphorus at 573 K in an inert atmosphere for several days.

59. Answer (4)

Hint: In Ostwald's process, NO formed combines with O_2 to give NO_2 which is dissolve in water to give HNO_3 .

Sol.: This process is based on catalytic oxidation of NH_3 .



60. Answer (2)

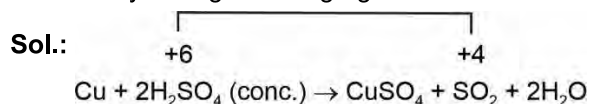
Hint: On moving down the group bond length increases $\text{B.E.} \propto \frac{1}{\text{Bond length}}$

Sol.:

Hydrides of group 15	Bond dissociation energy (kJ/mol)
NH_3	389
PH_3	322
AsH_3	297
SbH_3	255

61. Answer (3)

Hint: Hot concentrated sulphuric acid is a moderately strong oxidising agent.



62. Answer (4)

Hint: $\text{P}_4 + 3\text{NaOH} + 3\text{H}_2\text{O} \rightarrow \text{PH}_3 + 3\text{NaH}_2\text{PO}_2$
(sodium hypophosphite)

Sol.: • It gives phosphonium compounds with acids
 $\text{PH}_3 + \text{HBr} \rightarrow \text{PH}_4\text{Br}$

• It is a colourless gas with rotten fish smell

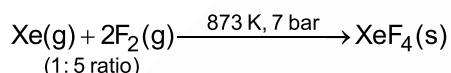
63. Answer (4)

Hint & Sol.:

Compound Name	Formula
Tear gas	CCl_3NO_2
Phosgene	COCl_2
Laughing gas	N_2O
Mustard gas	$\text{ClCH}_2\text{CH}_2\text{SCH}_2\text{CH}_2\text{Cl}$

64. Answer (2)

Hint: Xenon forms binary fluorides by the direct reaction of elements under appropriate experimental conditions.



Sol.:

65. Answer (1)

Hint & Sol.:

Formula	Physical state and colour
ClF_3	Colourless gas
BrF_3	Yellow green liquid
IF_3	Yellow powder
ICl_3	Orange solid

66. Answer (2)

Hint: When the polypeptide chains run parallel and are held together by hydrogen and disulphide bonds, then fibre-like structure is formed.

Sol.: Keratin and myosin are fibrous proteins.

67. Answer (3)

Hint & Sol.: Vitamin B1 is known as thiamine
Vitamin B2 is known as Riboflavin
Vitamin B6 is known as Pyridoxine
Vitamin C is known as Ascorbic acid

68. Answer (2)

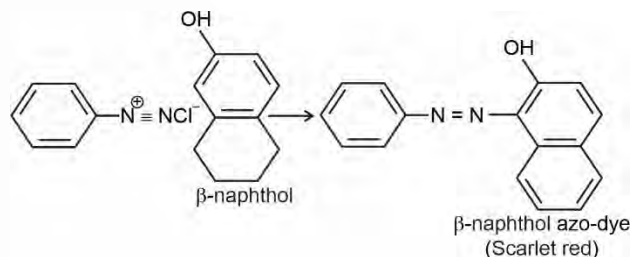
Hint: On oxidation with nitric acid, glucose as well as gluconic acid yield saccharic acid.



69. Answer (4)

Hint: On adding of β -naphthol to diazonium salt, coupling reaction take place to form a scarlet coloured dye which is sparingly soluble in water.

Sol.:



70. Answer (4)

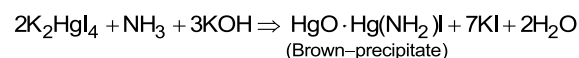
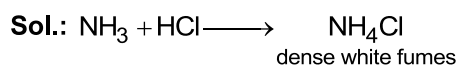
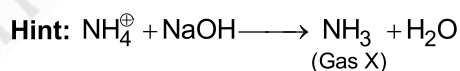
Hint & Sol.: AgI is yellow and insoluble in NH_3 solutions.

71. Answer (4)

Hint: H_2S is presence of NH_4OH is the group reagent for group IV.

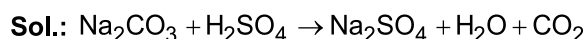
Sol.: When H_2S is passed from Zn salt solution in presence of NH_4OH , ZnS is formed, it is a white ppt.

72. Answer (1)



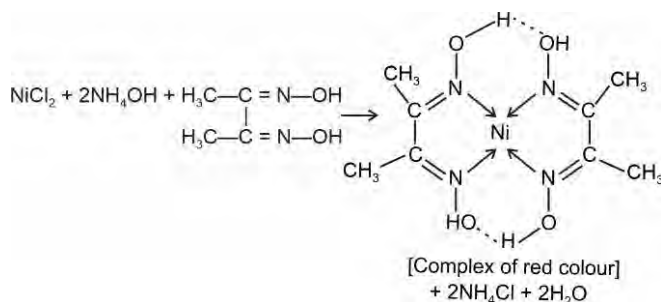
73. Answer (2)

Hint: Carbon dioxide is a colourless and odourless gas, which turns lime water milky.



74. Answer (2)

Hint: When dimethyl glyoxime is added to the aqueous solution of nickel chloride and is made alkaline by adding NH_4OH solution, a brilliant red precipitate is obtained.

Sol.:

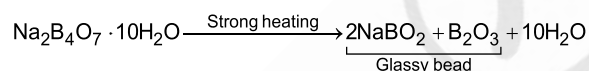
75. Answer (2)

Hint & Sol.: $\text{CuSO}_4 + \text{H}_2\text{S} \rightarrow \text{CuS} + \text{H}_2\text{SO}_4$
Black $\text{CuS} + 2\text{HNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + \text{H}_2\text{S}$
Blue solution $\text{Cu}(\text{NO}_3)_2 + 4\text{NH}_3 \rightarrow [\text{Cu}(\text{NH}_3)_4]^{2+} + 2\text{NO}_3^-$
Deep blue solution

76. Answer (2)

Hint & Sol.: Colloidal particle ranges from 1 nm to 1000 nm.

77. Answer (3)

Hint & Sol.:

78. Answer (3)

Hint: Molecular orbital diagram of S₂ is similar to that of O₂.**Sol.:** S₂ is paramagnetic with two unpaired electrons in π* orbitals.

79. Answer (3)

Hint: Percentage purity

$$= \frac{\text{Mass of pure chemical}}{\text{Mass of sample}} \times 100$$

Sol.: 20 ml of 0.1 N KMnO₄ correspond to = 0.002 eq. KMnO₄. Thus, 0.002 eq. ferrous ammonium sulphate is reacted. The eq. wt of ferrous ammonium sulphate is 392. 0.002 eq. of ferrous ammonium sulphate corresponds to

$$= 0.002 \times 392 = 0.784 \text{ g}$$

% purity of ferrous ammonium sulphate

$$= \frac{0.784}{1.96} \times 100 = 40\%$$

80. Answer (2)

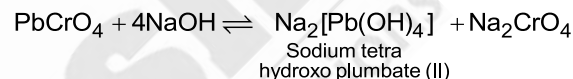
Hint: In halogens, higher member can displace lower member from their salt solution.**Sol.:** $2\text{KBr} + \text{I}_2 \rightarrow 2\text{KI} + \text{Br}_2$ Reaction is not possible because Br[⊖] ion is not oxidised in Br₂ with I₂ due to higher electrode potential of I₂ than bromine.

81. Answer (1)

Hint: s-block elements impart characteristic colour to Bunsen flame when heated on platinum loop.**Sol.:**

Element	Flame colour (naked eye)
Copper	— Green with blue centre
Barium	— Apple green
Strontium	— Crimson red
Calcium	— Brick red

82. Answer (4)

Hint: PbCrO₄ is insoluble in water and it forms a yellow precipitate.**Sol.:**

83. Answer (4)

Hint: Group reagents are used for the separation of cation due to their selective precipitation.**Sol.:**

Group	Group reagent
Group-I	Dilute HCl
Group-II	H ₂ S gas in presence of dil.HCl
Group-III	NH ₄ OH in presence of NH ₄ Cl
Group-IV	H ₂ S in presence of NH ₄ OH

84. Answer (4)

Hint: Methyl orange is a weak base and is yellow in colour in the unionised form.**Sol.:** Methyl orange is orange red in acidic medium and turns yellow in basic medium.

85. Answer (2)

Hint: On heating metal carbonates in a charcoal cavity, the carbonates are converted to metal oxide, which are coloured residues.**Sol.:** As³⁺ gives white residue with the odour of garlic.

86. Answer (2)

Hint & Sol.: Iodine forms intense blue complex with starch.

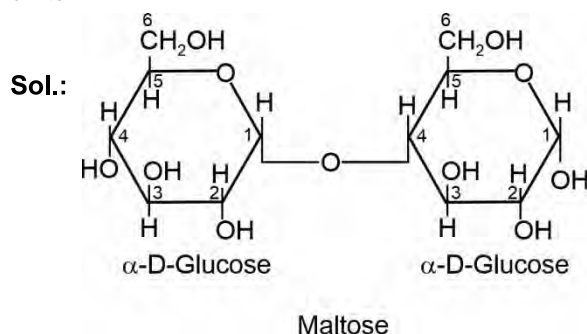
87. Answer (3)

Hint: In disaccharides, if the reducing groups of monosaccharides i.e., aldehydic or Ketonic groups are present in the form of acetal then they will not give positive Tollens' test and are called non-reducing sugar.

Sol.: In sucrose, the reducing groups of glucose and fructose are involved in glycosidic bond formation hence will not reduce Tollens' reagent.

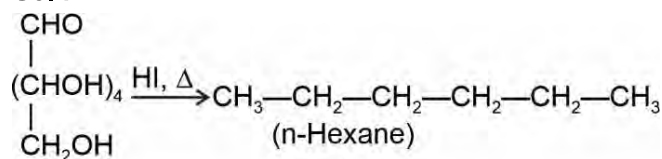
88. Answer (2)

Hint: Maltose is composed of two α -D-glucose units.



89. Answer (1)

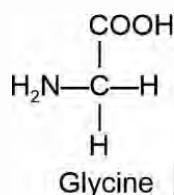
Hint: Glucose on prolong heating with HI, it forms n-hexane.

Sol.:

90. Answer (3)

Hint : Optically active compounds must have chiral carbon or molecular asymmetry.

Sol.: Glycine does not contain chiral centre.



[BOTANY]

91. Answer (2)

Hint: Sarguja, Chanda and Bastar are the places of sacred groves belonged to Madhya Pradesh.

Sol.: • Khasi and Jaintia Hills are in Meghalaya.

• Aravalli Hills are in Rajasthan.

• Western Ghat regions are in Karnataka and Maharashtra.

92. Answer (4)

Hint: Anthropogenic ecosystems are created and maintained by human being and they have little diversity.

Sol.: Anthropogenic ecosystem possesses high productivity with simple food chain.

93. Answer (3)

Hint: Key industry animals are first order consumers.

Sol.: Producers are also known as transducers because they convert solar energy into the chemical energy.

94. Answer (3)

Hint: Some of the key functional aspects of the ecosystem are productivity and decomposition.

Sol.: Stratification is the structural feature of the ecosystem.

95. Answer (4)

Hint: Primary productivity is the rate of biomass production by plants per unit area over a time period.

Sol.: Vertical stratification is more diverse in forests than in desert ecosystem as forests have more types of vegetation and form more different layers as compared to desert.

96. Answer (1)

Hint: Anthropogenic ecosystem is maintained by human beings.

Sol.: Estuaries (salt water) is a natural, aquatic ecosystem. Crop field, garden and aquarium are created by human beings.

97. Answer (4)

Hint: Species of birds (1300) are more in numbers as compared to mammals (427) in the Amazonian rain forests.

Sol.:

Taxa	Number of species
Mammals	427
Birds	1300
Reptiles	378
Amphibians	427
Fishes	3000

98. Answer (3)

Sol.: Primary productivity is expressed in terms of weight as $(\text{gm m}^{-2})\text{yr}^{-1}$ or energy as $(\text{kcal m}^{-2})\text{yr}^{-1}$.

99. Answer (3)

Hint: Temperature plays an important role in both latitudinal and altitudinal distribution of organisms.

Sol.: Mango, a tropical and subtropical tree plant is not found in temperate countries like Canada and Germany.

100. Answer (4)

Hint: Euryhaline are organisms which tolerate a wide range of salinities.

Sol.: Stenohaline are those organisms, which are restricted to a narrow range of salinities.

101. Answer (1)

Hint: Warm-blooded animals (birds and mammals) have larger body size in cold climates than in hotter areas. This is called Bergmann's rule.

Sol.: Mammals from colder climates generally have shorter ears and limbs to minimise heat loss. This is called as Allen's rule.

• **Rensch's rule:** In colder climate, birds possess narrow and acuminate wings as compared to broader wings of birds found in warmer areas.

• **Jordan's rule:** As the temperature is lowered, some fishes possess larger size with larger number of vertebrae.

102. Answer (1)

Hint: Emigration is the number of individuals of the population who had left the habitat and gone elsewhere during the time period under consideration.

Sol.: Natality refers to the number of births during a given period in the population that are added to initial density.

• Mortality is the number of deaths in the population during given period.

• Immigration is the number of individuals of the same species that have come into the habitat from elsewhere during time period under consideration.

103. Answer (4)

Hint: Sex ratio is an important population attribute.

Sol.: Birth rate and death rates are important population attributes.

An individual may have birth and death.

104. Answer (4)

Hint: In predation, only one species (predator) benefits and the interaction is detrimental to the other species (Prey).

Sol.: In competition, both the interacting species are negatively affected.

105. Answer (1)

Hint: Mineralisation is the release of inorganic substances from organic matter.

Sol.: Standing crop is the amount of all the living material present in an ecosystem per unit area at a given time.

• Standing state or quality is the amount of all the inorganic substances present in an ecosystem per unit area at a given time.

106. Answer (3)

Hint: Ecosystem diversity is high in India, because of the occurrence of large number of ecosystems like desert, rain forests and many rivers etc.

Sol.: Species diversity is the product of species richness and species evenness.

107. Answer (3)

Hint: An urn-shaped age pyramid shows declining population.

Sol.: Urn-shaped age pyramid shows higher number of reproductive individuals as compared to pre-reproductive individuals.

108. Answer (3)

Hint: In species-area relationship equation, species richness (S) is directly proportional to the area (A).

Sol.: The correct equation of species-area relationship on a logarithmic scale is

$$\log S = \log C + Z \log A$$

109. Answer (4)

Hint: The hotspots are richest and most threatened reservoirs of plants and animals life on earth.

Sol.: Collection of forest products, harvesting of timber and tilling of land are allowed in sanctuaries. Drugs and industrial products are included in narrowly utilitarian categories while aesthetic pleasure is included in broadly utilitarian categories.

110. Answer (4)

Hint: Humans have always been dependent on nature for food and shelter, but when need turns to greed it leads to overexploitation of natural resources.

Sol.: Habitat loss and fragmentation is the most important cause driving animals and plants to extinction.

111. Answer (2)

Sol.: The World Summit on Sustainable Development was held in 2002 in Johannesburg, South Africa.

112. Answer (4)

Hint: Dodo was extinct from Mauritius.

Sol.: Steller's sea cow was the most recent extinction from Russia.

113. Answer (1)

Hint: Presence of thorns are protective mechanisms shown by plants.

Sol.: Plants can avoid their predators like phytophagous insects because plants have evolved an astonishing variety of morphological and chemical defence against herbivores.

114. Answer (3)

Hint: Bear escapes in time during winter by a process caused hibernation.

Sol.: A stage of suspended development found in many zooplankton species is called diapause.

115. Answer (4)

Hint: In accordance with life styles, parasites have evolved a number of special adaptations.

Sol.: Parasites show high reproductive capacity, loss of digestive system, etc. as adaptations to survive in host.

116. Answer (2)

Hint: Stratification is the vertical distribution of different species occupying different levels.

Sol.: Species composition and stratification are the structural components of an ecosystem.

117. Answer (3)

Hint: Mycorrhiza is a mutualistic relationship between fungi and roots of higher plants.

Sol.: Commensalism – orchids growing on mango tree.

Parasitism – Ticks on dogs.

118. Answer (3)

Hint: Ecological pyramids assume a simple food chain and does not accommodate a food web.

Sol.: Saprophytes, decomposers, microbes and detritivores are not given any place in ecological pyramids.

119. Answer (2)

Hint: Mineralisation is the release of inorganic substances from organic matter.

Sol.: Humification is the process of decomposition of detritus to form humus. Humus is a dark coloured, amorphous, more or less decomposed organic matter rich in cellulose, lignin etc.

In leaching, part of water-soluble substances present in the fragmented and decomposing detritus go down into the soil horizon by percolating water and get precipitated as unavailable salts.

120. Answer (3)

Hint: Very less amount of energy is present at highest trophic level.

Sol.: Pyramid of energy is always upright in all ecosystem. Because only 10% energy is transferred from one trophic level to another trophic level and remaining 90% is consumed in respiration and other metabolic activities.

121. Answer (3)

Hint: The transfer of energy from one trophic level to another trophic level is accompanied by loss of energy at each level or step.

Sol.: Only 10% energy is transferred from one trophic level to another trophic level.

$$\begin{array}{ccccccc} T_1 & \longrightarrow & T_2 & \longrightarrow & T_3 & \longrightarrow & T_4 \\ 5000 \text{ J} & & 500 \text{ J} & & 50 \text{ J} & & 5 \text{ J} \\ \text{NPP} & & & & & & \end{array}$$

So, the amount of energy which gets stored and fixed in tertiary consumer will maximally be 5 J.

122. Answer (2)

Hint: A Biosphere reserve consists of core, buffer and transition zones.

Sol.: The buffer zone surrounds the core area and is managed to accommodate a greater research and educational activities.

123. Answer (1)

Hint: Rate of decomposition of detritus is controlled by chemical nature of detritus and number of climatic factors.

Sol.: Temperature and soil moisture are the most important climatic factors that regulate decomposition through their effect on the activities of soil microbes.

An optimum moisture helps in quicker decomposition of detritus.

At a temperature of more than 25°C, decomposers are very active in soils having good moisture and aeration.

124. Answer (3)

Hint: Rabbit is herbivore and it is primary consumer in grassland ecosystem.

Sol.: Grass → Grasshopper → Frog → Snake → Eagle.

125. Answer (1)

Hint: The effect of reduction in biodiversity has been explained by Paul Ehrlich through Rivet popper hypothesis.

Sol.: • Species-area relationships was given by Alexander von Humboldt.

• Term biodiversity was popularised by Edward Wilson.

• According to Robert May, the number of global species diversity is about 7 million.

126. Answer (4)

Sol.: Sparrow occupies more than one trophic level in the same ecosystem at the same time.

127. Answer (4)

Hint: Energy flow in an ecosystem is unidirectional.

Sol.: Energy flow follows laws of thermodynamics.

128. Answer (1)

Hint: Over exploitation of particular species reduces size of its population to an extent so that it becomes vulnerable to extinction.

Sol.: New species entering a geographical region are called exotic or alien or non-native species e.g., Nile perch, was introduced into Lake Victoria of East Africa.

129. Answer (3)

Hint: Archaeobacteria sustain their life under extremes of temperature due to branched chain lipid in membrane.

Sol.: Bear escape in time during winter by a process called hibernation.

130. Answer (4)

Sol.: By these population characters, the density of population (N) at time 't' can be expressed after a period of time 't + 1'

$$\text{As } N_{t+1} = N_t + [(B + I) - (D + E)]$$

$$B = 350$$

$$I = 20$$

$$D = 340$$

$$E = 30$$

$$N_{t+1} = N_t + [(350 + 20) - (340 + 30)]$$

$$N_{t+1} = \text{Zero}$$

131. Answer (2)

Hint: Any species growing exponentially under unlimited resource conditions can reach enormous population densities in a short time.

Sol.: The integral form of exponential growth equation will be

$$N_t = N_0 e^{rt}$$

N_t = population density after 't' time

N_0 = population density at time zero

e = The base of natural logarithms.

132. Answer (4)

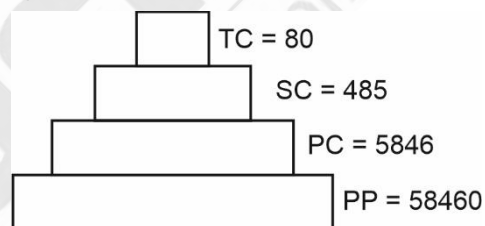
Hint: Decomposers are also called reducers as they are capable of degrading the dead organism.

Sol.: Decomposers secrete digestive enzymes which convert complex organic substances into simpler ones.

133. Answer (3)

Hint: Pyramid of number in aquatic ecosystem can be upright.

Sol.:



134. Answer (3)

Hint: Predators help in maintaining species diversity in a community.

Sol.: Biological pest control methods used in agriculture are based on the ability of predator to regulate prey population.

135. Answer (3)

Hint: Communities with more species tend to be more stable.

Sol.: A stable community should not show too much variation in productivity from year to year.

[ZOOLOGY]

136. Answer (2)

Hint: Also constructed artificial recombinant DNA for the first time.

Sol.: Stanley Cohen developed a method of removing plasmids from the cell and then reinserting them in other cells.

Herbert Boyer observed about restriction enzymes of the *E.coli* and their useful properties.

S.L. Miller did a simulation experiment for chemosynthesis.

Watson and Crick gave double helical model of B-DNA.

137. Answer (2)

Hint: Upkeeps the normal standards of GM research

Sol.: Genetic Engineering Approval Committee has been setup by Indian government to make decisions regarding the validity of GM research and safety of introducing GM-organisms or products for public services.

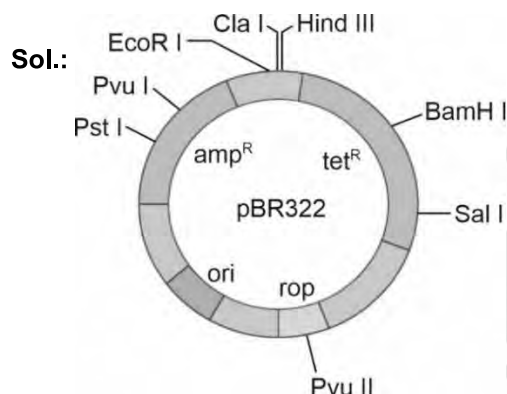
NACO → National AIDS Control Organisation

EFB → European Federation of Biotechnology

WHO → World Health Organisation.

138. Answer (3)

Hint: Exclude the enzymes whose restriction site is within tet^R



139. Answer (3)

Hint: DNA ligase

Sol.: DNA ligases also known as molecular glue belong to class VI of enzymes, catalyse the linking together of alien DNA and cloning vector.

Restriction endonucleases also known as molecular scissors help in cutting the DNA at specific locations.

140. Answer (3)

Hint: Endonucleases are found in prokaryotes.

Sol.: First letter of the name of restriction enzyme for e.g., *Bam*HI comes from the genus which is *Bacillus*, second two letters comes from species which is *amyloliquefaciens*.

The letter 'H' is derived from the strain. Roman numbers following the name which is 'I' indicates the order in which enzyme was isolated from that strain of bacteria.

141. Answer (1)

Hint: Series of processes that also include marketing of the product

Sol.: Transformation is a procedure through which a piece of DNA is introduced in a host cell.

Gel electrophoresis → Technique utilised for separation of DNA fragments after digestion by endonucleases.

Upstream processing → Includes all the processes upto formation of recombinant protein.

142. Answer (2)

Hint: Insertional inactivation

Sol.: The ligation of foreign DNA at *Eco*RI site will not lead to any insertional inactivation of the antibiotic resistance gene. Due to which recombinant *E.coli* will be resistant against both tetracycline and ampicillin antibiotics.

143. Answer (3)

Hint: Include the genetic disorder

Sol.: Today, transgenic models exist for human diseases such as cancer, Alzheimer's, cystic fibrosis and Rheumatoid arthritis.

144. Answer (4)

Hint: Conventional method of diagnosis

Sol.: Recombinant DNA technology, Polymerase Chain Reaction and Enzyme Linked Immunosorbent Assay are some of the techniques that serve the purpose of early diagnosis.

Urine analysis and serum analysis are conventional method of diagnosis.

145. Answer (1)

Hint: Use of agrochemicals.

Sol.: The Green Revolution succeeded in tripling the food supply but yet it was not enough to feed the growing human population. Increased yields have partly been due to the use of improved crop varieties, but mainly due to the use of better management practices and use of agrochemicals (fertilisers and pesticides).

146. Answer (3)

Hint: Exclude the Bt crop.

Sol.: In 1997, American company got patent over Basmati rice and same attempts have been made to patent a, product and process based on Indian traditional herbal medicine such as neem and turmeric.

147. Answer (4)

Hint: Mature insulin has one intrachain disulfide bond

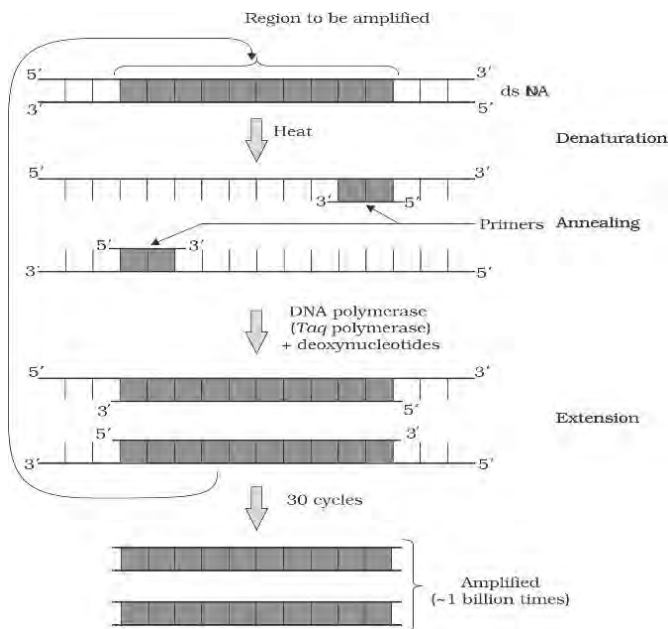
Sol.: Insulin used for diabetes earlier was extracted from pancreas of slaughtered cattle and pigs. Insulin is synthesised as a pro-hormone and before it becomes fully mature, an extra stretch called C peptide is removed and mature form contains two short polypeptides called A and B.

In 1983, Eli Lilly, an American company, prepared two DNA sequences corresponding to A and B chains of human insulin and introduced them in plasmids of *E.coli* to produce human insulin.

148. Answer (3)

Hint: Used for animal cells**Sol.:** In vectorless method known as micro-injection, recombinant DNA is directly injected into the nucleus of an animal cell.

149. Answer (3)

Hint: Guanylic acid is a nucleotide.**Sol.:**

Guanosine is a nucleoside and cytosine is a nitrogenous base.

150. Answer (2)

Hint: Insertional inactivation

Sol.: β -galactosidase produce blue colour in presence of chromogenic substrate. Now, if foreign DNA is inserted within β -galactosidase coding sequence, it will lead to insertional inactivation due to which recombinants will produce white coloured colonies and non-recombinants will produce blue coloured colonies in presence of chromogenic substrate.

151. Answer (4)

Hint: Used as a selectable marker.**Sol.:**

Transposons	–	Mobile genetic elements
Agarose	–	Natural polymer extracted from sea weeds, used as matrix in gel electrophoresis
Lysozyme	–	Enzyme used to breakdown bacterial cell wall
Chloramphenicol	–	Antibiotic

152. Answer (1)

Hint: Spooling

Sol.: DNA isolation can be achieved by treating the bacterial cells / plant or animal tissue with enzymes such as lysozyme (bacterial), cellulase (plant) or chitinase (fungus). DNA ligase assists in ligation of fragments of DNA.

Ribonuclease can be used to remove RNA. Purified DNA ultimately precipitates out after the addition of chilled ethanol and can be seen as collection of fine threads in the suspension.

153. Answer (4)

Hint: More than the normal human body temperature**Sol.:** Heat shock for bacterial transformation is done at 42°C.

Denaturation of nucleic acids in PCR is done at 94°C.

Thermophilic enzymes remain stabilised upto 80-90°C and normal human body temperature stays in the range of 36-37°C.

154. Answer (1)

Hint: Continuous culture system

Sol.: The initial phase is the lag phase where bacteria are metabolically active but not dividing. The exponential or log phase is the time of exponential growth. To obtain large culture, cells are kept in log phase.

In the stationary phase, growth reaches a plateau.

In death or decline phase, growth of bacteria decreases again.

155. Answer (2)

Hint: Exclude the events in 1990s

Sol.: In the year 1963, the two enzymes responsible for restricting the growth of bacteriophage in *Escherichia coli* were isolated. One of those enzyme was restriction endonuclease. The first restriction endonuclease *HindII* was isolated and characterised five years later.

In 1990, first clinical gene therapy was given to a 4-years old girl.

In 1997, first transgenic cow, Rosie was produced.

156. Answer (3)

Hint: Methylase and restriction endonuclease

Sol.: In 1963, two enzymes were discovered for restricting the growing of bacteriophage in *E.coli*. One (methylase) added methyl groups to DNA of bacteria and the other (endonuclease) cuts the DNA of bacteriophage.

157. Answer (2)

Hint: Restriction enzymes are nucleases**Sol.:** Nucleases are of two types : exonucleases and endonucleases. Exonucleases remove nucleotides from the ends of the DNA whereas endonucleases make cuts at specific positions within the DNA.

Restriction endonucleases are naturally present in prokaryotic cells.

158. Answer (4)

Hint: Formation of sticky ends**Sol.:** Restriction enzymes cut the strand of DNA little away from centre of the palindromic sequence and leave sticky ends on each strand. The stickiness of ends facilitates the action of enzyme DNA ligase when both alien DNA and plasmid are cut by same restriction enzyme.

159. Answer (2)

Hint: Belongs to the order Lepidoptera**Sol.:** Bt toxins are insect-group specific.Proteins encoded by the genes *cryIAC* and *cryII Ab* control the cotton bollworms and that of *cry IAb* controls corn borer.

160. Answer (2)

Hint: Treatment of emphysema**Sol.:** Insulin is given for treatment of diabetes. α -1-antitrypsin is given to treat emphysema. α -lactalbumin is present in the milk produced by first transgenic cow, Rosie.

161. Answer (2)

Hint: Steps in autoradiography**Sol.:** A single stranded DNA or RNA, tagged with a radioactive molecule (probe) is allowed to hybridised to its complementary DNA in a clone of cells followed by detection using radiography. The clone having the mutated gene will hence not appear on the photographic film, because the probe will not have complementarity with the mutated gene.

162. Answer (2)

Hint: Safety testing of polio vaccine**Sol.:** Transgenic mice are being developed for use in testing the safety of vaccines before they are used on humans. Transgenic mice are being used to test the safety of the polio vaccine. If successful and found to be reliable, they could replace the use of monkeys to test the safety of batches of vaccine.

163. Answer (4)

Hint: *Agrobacterium tumefaciens* possesses Ti plasmid**Sol.:** *Agrobacterium tumefaciens*, a pathogen of several dicot plants is able to deliver a piece of DNA known as 'T-DNA' to transform normal plant cells into a tumor.Ti plasmid (Tumor inducing plasmid of *Agrobacterium*) has been modified into a cloning vector which is no more pathogenic to the plants.*Meloidogyne* is a nematode which infects tobacco plants.

164. Answer (2)

Hint: Exclude the treatment after birth**Sol.:** Children suffering from ADA deficiency can be cured by bone marrow transplantation, enzyme replacement therapy, gene therapy but problem with all these methods is that they are not completely curative in nature.

However, if the gene isolated from marrow cells producing ADA is introduced into cells at early embryonic stages then it could be a permanent cure.

165. Answer (3)

Hint: Insecticidal protein**Sol.:** Some strains of *Bacillus thuringiensis* produce proteins that kill certain insects such as lepidopterans (Tobacco budworm, Armyworm), coleopterans (Beetles) and dipterans (Flies, Mosquitoes).*Agrobacterium* is a pathogen which causes crown gall disease in dicots.*Meloidogyne incognita* is a nematode which infects the roots of tobacco plants.

Disarmed retrovirus can be used as vector to deliver foreign DNA in an animal cell.

166. Answer (1)

Hint: Insulin consists of peptides linked together by disulfide bridges.**Sol.:**

GM plants	–	Fuels and pharmaceuticals
ADA deficiency	–	Genetically engineered lymphocytes
Mature insulin	–	Two interchain disulphide bonds present in between chain A and B and one intrachain disulphide bond present in chain A

167. Answer (2)

Hint: RNA interference**Sol.:** RNAi involves silencing of a specific mRNA due to complementary dsRNA molecule that binds to and prevents translation of the mRNA (silencing).

168. Answer (1)

Hint: Choose the order to which cotton bollworm belongs to**Sol.:**

Lepidopterans	–	Tobacco budworm, armyworm
Coelopterans	–	Beetles
Dipterans	–	Flies, mosquitoes

169. Answer (3)

Hint: Used for separation and isolation of DNA fragments.**Sol.:** DNA fragments are negatively charged and can be separated by forcing them to move towards the anode under an electric field through a medium. DNA fragments separate according to their size under sieving effect by agarose gel. Separate bands of DNA are cut out and extracted from the gel piece by elution.

170. Answer (1)

Hint: Facilitate the mixing of the reactor contents.**Sol.:** Stirred tank bioreactor contains flat bladed impeller, curved base with foam breaker to facilitate the mixing of the reactor contents.

When protein encoding gene is expressed in heterologous host, protein produced is called recombinant protein.

First recombinant DNA was constructed by linking a gene encoding the antibiotic resistance with native plasmid of *Salmonella typhimurium*.

171. Answer (2)

Hint: Reporter enzyme**Sol.:** Selection of recombinants due to inactivation of antibiotics is a cumbersome procedure. Therefore, alternative selectable markers have been developed which differentiate recombinants from non-recombinants on the basis of their ability to produce colour in the presence of a chromogenic substrate. A recombinant DNA is inserted within the coding sequence of an enzyme, β -galactosidase.Reporter enzyme such as β -galactosidase can be used to monitor transformation of host cells.

172. Answer (3)

Hint: Gene gun**Sol.:** In plants, cells are bombarded with high velocity microparticles of gold or tungsten coated with DNA in a method known as biolistics or gene gun. Microinjection is used for animals such as *Rattus*, nematodes and *Meloidogyne*.

173. Answer (3)

Hint: Can be used for treating night blindness**Sol.:** Genetic modification in plants are useful as they

- (i) Made crops more tolerant to abiotic stresses
- (ii) Reduced reliance on chemical pesticides.
- (iii) Enhanced nutritional value of food, e.g. Golden rice, i.e. Vitamin 'A' enriched rice.

174. Answer (1)

Hint: Tissue culture**Sol.:** If the plant is infected with a virus, the meristem (apical and axillary) is free of virus. Hence, one can remove the meristem and grow it *in vitro* to obtain virus free plants.

175. Answer (4)

Hint: Exclude the nematode**Sol.:** Pomato is not a GMO but formed by somatic hybridisation but Rosie is a GMO.*Meloidogyne* – Nematode*Bacillus thuringiensis* – Microbe

176. Answer (2)

Hint: Processes of rDNA**Sol.:** Recombinant DNA technology involves several steps in specific sequence such as isolation of DNA, fragmentation of DNA by endonuclease, ligation of the DNA fragment into a vector, transferring the recombinant DNA into the host, culturing the host cells in a medium at large scale and extraction of the desired product.

177. Answer (1)

Hint: Features that are required to facilitate cloning**Sol.:** *ori* sequence is responsible for controlling the copy number of the linked DNA.Normal *E.coli* cells do not carry resistance against ampicillin and tetracycline antibiotics. Vectors need to have very few, preferably single restriction site for commonly used restriction enzymes. Presence of more than one recognition sites within the vector will generate several fragments which will complicate the gene cloning.

178. Answer (2)

Hint: Use of only natural resources

Sol.: The applications of biotechnology include the early diagnosis, genetically modified crops for agriculture, processed food, bioremediation, waste treatment.

Traditional hybridisation procedures used in plant and animal breeding, very often lead to inclusion and multiplication of undesirable genes along with the desired genes.

179. Answer (2)

Hint: Part of chemical engineering

Sol.: Two techniques enabled the birth of modern biotechnology.

a. Genetic engineering → Techniques that alter the chemistry of genetic material (DNA and RNA) to introduce these into host organisms and change the phenotype of plant / animal.

b. Bioprocess engineering → Maintenance of the sterile environment for mass production of biotechnological products.

180. Answer (1)

Hint: Bacterial transformation

Sol.: In order to force bacteria to take up the plasmid, the bacterial cells are made competent by treating them with the specific concentration of a divalent cation like Ca^{2+} . DNA is a negatively charged molecule.



All India Aakash Test Series for NEET - 2025

TEST - 7 (Code-F)**Click here for
Code-E sol.**

Test Date : 23/03/2025

ANSWERS

1. (3)	37. (4)	73. (4)	109. (4)	145. (2)
2. (4)	38. (3)	74. (4)	110. (4)	146. (1)
3. (2)	39. (3)	75. (3)	111. (3)	147. (3)
4. (1)	40. (4)	76. (2)	112. (3)	148. (1)
5. (4)	41. (4)	77. (4)	113. (3)	149. (2)
6. (4)	42. (1)	78. (2)	114. (4)	150. (1)
7. (1)	43. (3)	79. (4)	115. (2)	151. (3)
8. (3)	44. (2)	80. (4)	116. (4)	152. (2)
9. (3)	45. (4)	81. (3)	117. (3)	153. (4)
10. (3)	46. (3)	82. (1)	118. (1)	154. (2)
11. (2)	47. (1)	83. (1)	119. (4)	155. (2)
12. (2)	48. (2)	84. (3)	120. (4)	156. (2)
13. (4)	49. (3)	85. (4)	121. (1)	157. (2)
14. (2)	50. (2)	86. (2)	122. (3)	158. (4)
15. (1)	51. (2)	87. (1)	123. (1)	159. (2)
16. (3)	52. (4)	88. (2)	124. (2)	160. (3)
17. (2)	53. (4)	89. (1)	125. (3)	161. (2)
18. (1)	54. (4)	90. (4)	126. (3)	162. (1)
19. (1)	55. (1)	91. (2)	127. (2)	163. (4)
20. (3)	56. (2)	92. (4)	128. (3)	164. (1)
21. (4)	57. (3)	93. (3)	129. (3)	165. (4)
22. (3)	58. (3)	94. (3)	130. (2)	166. (2)
23. (1)	59. (3)	95. (4)	131. (4)	167. (3)
24. (4)	60. (2)	96. (1)	132. (3)	168. (3)
25. (2)	61. (2)	97. (4)	133. (1)	169. (4)
26. (3)	62. (2)	98. (3)	134. (4)	170. (3)
27. (1)	63. (2)	99. (3)	135. (2)	171. (1)
28. (4)	64. (1)	100. (4)	136. (1)	172. (4)
29. (1)	65. (4)	101. (1)	137. (2)	173. (3)
30. (2)	66. (4)	102. (1)	138. (2)	174. (2)
31. (4)	67. (4)	103. (4)	139. (1)	175. (1)
32. (4)	68. (2)	104. (4)	140. (2)	176. (3)
33. (1)	69. (3)	105. (1)	141. (4)	177. (3)
34. (2)	70. (2)	106. (3)	142. (1)	178. (3)
35. (3)	71. (1)	107. (3)	143. (3)	179. (2)
36. (4)	72. (2)	108. (3)	144. (3)	180. (2)

HINTS & SOLUTIONS

[PHYSICS]

1. Answer (3)

Hint & Sol.: In forward biasing, p -side of diode is connected to positive terminal of battery and n -side is connected to negative terminal of battery. In forward biasing, width of depletion layer decreases.

2ss. Answer (4)

Hint: $\sin C = \frac{1}{\mu} = \frac{1}{\sqrt{3}}$ where C is critical angle of face AC .

Sol.: $i = 60^\circ$ on face AC .

$$\sin i > \sin C \quad \left(\because \frac{\sqrt{3}}{2} > \frac{1}{\sqrt{3}} \right)$$

$$\Rightarrow i > C$$

Ray undergoes TIR on face AC .

Angle of emergence doesn't exist.

3. Answer (2)

Hint: $2\pi r = n\lambda$ and $r_n = \frac{0.529n^2 \text{ \AA}}{z}$

Sol.: $n = 2$ (first excited state)

$$z = 1$$

$$\lambda = \pi r_2$$

$$\lambda = \pi(0.529 \text{ \AA}) \frac{(2^2)}{1}$$

$$\lambda = 2.12\pi \text{ \AA}$$

4. Answer (1)

Hint : Use, $eV_0 = h\nu - \phi_0$

$$\text{Sol. : } V_0 = \frac{h}{e} \nu - \frac{\phi_0}{e}$$

$$\therefore \text{ slope } m = \frac{h}{e}$$

5. Answer (4)

Hint & Sol.: Energy required to remove an e^- from ground state of hydrogen atom is 13.6 eV

$$\therefore r = r_0 = 0.53 \text{ \AA}$$

6. Answer (4)

Hint & Sol.: Large line source will produce cylindrical wavefront.

7. Answer (1)

$$\text{Hint: } \frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\text{Sol.: } \frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{3}{2v} + \frac{1}{15} = \frac{0.5}{-30}$$

$$\Rightarrow \frac{3}{2v} = \frac{1}{-60} - \frac{4}{60} = \frac{-5}{60}$$

$$v = -18 \text{ cm (virtual image)}$$

8. Answer (3)

Hint: For a metre bridge at null point $\frac{P}{Q} = \frac{l}{100-l}$

$$\text{Sol.: } \frac{P}{Q} = \frac{60}{40}$$

$$\frac{6}{Q} = \frac{60}{40}$$

$$Q = 4 \Omega$$

9. Answer (3)

$$\text{Hint: } \frac{P}{Q} = \frac{l}{100-l}$$

$$\text{Sol.: } \frac{6}{3} = \frac{2}{1} = \frac{l}{100-l}$$

$$\text{Null point} = 66.7 \text{ cm}$$

$$\Rightarrow \text{Shift} = 6.7 \text{ cm}$$

10. Answer (3)

Hint: Use Einstein's photoelectric equation.

$$\text{Sol.: } k_1 = h\nu - \phi$$

$$k_2 = 2h\nu - \phi$$

$$k_2 = 2(k_1 + \phi) - \phi$$

$$= 2k_1 + \phi$$

$$\frac{1}{2} M v_2^2 = 2 \times \frac{1}{2} \times M v_1^2 + \phi$$

$$v_2 = \sqrt{2v_1^2 + \frac{2\phi}{M}}$$

11. Answer (2)

Hint: Fringe width $\beta = \frac{\lambda D}{d}$

Sol.: $\frac{\beta_2}{\beta_1} = \frac{\lambda_2}{\lambda_1} \Rightarrow \lambda_2 = 560 \text{ nm}$

12. Answer (2)

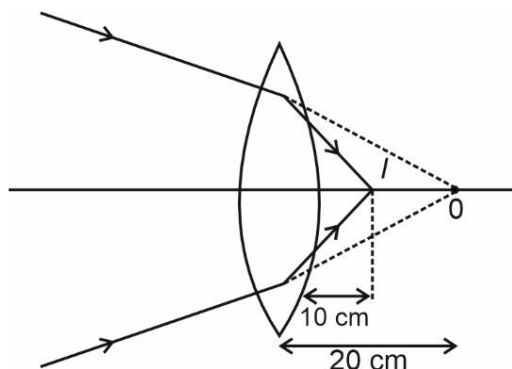
Hint & Sol.: Searle's apparatus is used for the measurement of Young's modulus.

13. Answer (4)

Hint & Sol.: Reflection, interference and diffraction can all be explained by wave phenomena.

14. Answer (2)

Hint & Sol.:



By lens formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{+10} - \frac{1}{20} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{20} = \frac{1}{f}$$

$$\Rightarrow f = 20 \text{ cm}$$

15. Answer (1)

Hint: $E_n = \frac{-13.6Z^2}{n^2} \text{ eV}$

Sol.: $E_3 - E_1$ energy difference is greatest among the given options in the question.

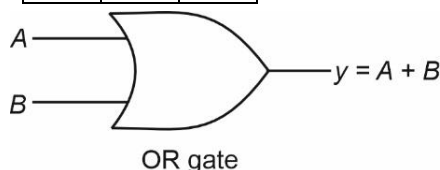
$\therefore$ In case-I, most energetic photon will emit.

16. Answer (3)

Hint: When both A and B are open then we cannot get any output.

Sol.:

A	B	C
0	0	0
0	1	1
1	0	1
1	1	1



17. Answer (2)

Hint & Sol.: Nuclides with same neutron number N but different atomic number Z are, isotones.

18. Answer (1)

Hint: Use Snell's law

$$n_1 \sin i = n_2 \sin r$$

Sol.: $n_1 \sin i = n_2 \sin r_1$

$$n_2 \sin r_1 = n_3 \sin r_2$$

$$n_3 \sin r_2 = n_4 \sin r_3$$

$$n_4 \sin r_3 = n_5 \sin e$$

Comparing all equations

$$n_1 \sin i = n_5 \sin e$$

If $n_1 = n_5$ then incident ray and emergent ray are parallel to each other.

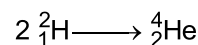
19. Answer (1)

Hint: An intrinsic semiconductor will behave like an insulator at $T = 0 \text{ K}$ and at $T > 0 \text{ K}$, some thermally generated electron hole pairs are to be depicted.

Sol.: It is the thermal energy at higher temperatures ($T > 0 \text{ K}$), which excites some electrons from the valence band to conduction band. These thermally excited electrons at $T > 0 \text{ K}$, partially occupy the conduction band. Therefore, the energy-band diagram should be option (1)

20. Answer (3)

Hint: Given reaction represented as



Sol.: Energy released (Q-value)

$$= (B.E.)_{\text{product}} - (B.E.)_{\text{reactant}}$$

$$= b - 2a$$

21. Answer (4)

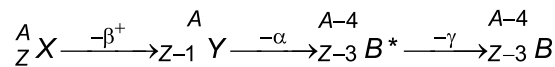
Hint & Sol.: For any semiconductor: $n_i^2 = n_e n_h$

22. Answer (3)

Hint: In β^+ decay, atomic number decreases by 1 and mass number remains same.

In decay process when product nucleus is in its excited state, then it emits a γ -photon (γ -decay)

Sol.:



23. Answer (1)

Hint: Use formula $R = R_0 A^{1/3}$

Sol.: $R = R_0 A^{1/3}$

$$= 1.2 \times (70)^{1/3}$$

$$= 4.94 \text{ fm}$$

24. Answer (4)

Hint & Sol.: Properties of intrinsic semiconductor.

25. Answer (2)

Hint: $m = \frac{f_0}{f_e}$

Sol.: $m = \frac{100}{1} = 100$

26. Answer (3)

Hint & Sol.: The minimum energy to ionise an atom is the energy required to remove the most loosely bound (outermost) electron.

27. Answer (1)

Hint: P.E. of hydrogen atom in ground state is -27.2 eV .

Sol.: If the P.E. in ground state is taken as zero then whole system is shifted by 27.2 eV . So, T.E. in first excited state will be

$$(-3.4 + 27.2) \text{ eV}$$

$$= 23.8 \text{ eV}$$

28. Answer (4)

Hint: Properties of semiconductors.

Sol.: For insulators $E_g > 3 \text{ eV}$, for semiconductor E_g is 0.2 eV to 3 eV while for metals $E_g \approx 0$

29. Answer (1)

Hint: $\frac{1}{\lambda} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$

Sol.: For shortest wavelength of Lyman series:

$$n_1 = 1, n_2 = \infty$$

$$\frac{1}{\lambda} = R$$

$$\lambda = \frac{1}{R}$$

30. Answer (2)

Hint: P.E. = $\frac{-kZe^2}{r}$

T.E. = $\frac{\text{P.E.}}{2}$

Sol.: As principal quantum number (n) increases, the radius of the orbit increases and potential energy becomes less negative (increases). As n increases, T.E. also increases

31. Answer (4)

Hint & Sol.: Between two nucleons, the nuclear force is attractive for separation larger than 0.8 fm . The nuclear mass is always less than the total mass of its constituents.

32. Answer (4)

Hint & Sol.: In explosion $\Rightarrow$ Momentum remain conserved

$$\lambda = \frac{h}{p}$$

$$\text{Since } p_1 = p_2 \Rightarrow \lambda_1 = \lambda_2$$

33. Answer (1)

Hint & Sol.: As temperature rises, the atoms of the liquid become more mobile and the coefficient of viscosity η falls. In a gas as temperature rises, it increases the random motion of atoms and as a consequence viscosity of gases increases.

34. Answer (2)

Hint: Use Einstein's equation of photoelectric effect, $K_{\max} = h\nu - \phi$

Sol.: $1.2 = h\nu - \phi$ (i)

$3.6 = 1.5 h\nu - \phi$ (ii)

On solving equation (i) and (ii)

$$\therefore \phi = 3.6 \text{ eV}$$

35. Answer (3)

Hint & Sol.: For a given frequency of the incident radiation, the stopping potential is independent of its intensity. The maximum kinetic energy of photoelectrons depends on frequency of the light source and the emitter plate material, but is independent of intensity of incident radiation.

36. Answer (4)

Hint: When an object is placed at the second focus of a concave lens, the image will be formed at infinity.

Sol.: For first lens, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} - \frac{1}{-15} = \frac{1}{10}$

$$\Rightarrow v = 30 \text{ cm}$$

Virtual object for second lens at 25 cm from this lens,

$$\frac{1}{v'} - \frac{1}{25} = \frac{-1}{25} \Rightarrow v' = \infty$$

37. Answer (4)

$$\text{Hint: } I = 4I_0 \cos^2\left(\frac{\Delta\phi}{2}\right)$$

$$\text{Sol.: If } \Delta x = \frac{\lambda}{3}, \Delta\phi = \frac{2\pi}{3}, I = 4I_0 \cos^2 60^\circ = I_0$$

$$\text{If } \Delta x = \frac{\lambda}{6}, \Delta\phi = \frac{\pi}{3}, I = 4I_0 \cos^2 30^\circ = 3I_0$$

$$\text{If } \Delta x = \frac{\lambda}{4}, \Delta\phi = \frac{\pi}{2}, I = 4I_0 \cos^2 \frac{\pi}{4} = 2I_0$$

$$\text{If } \Delta x = \frac{\lambda}{2}, \Delta\phi = \pi, I = 4I_0 \cos^2 \frac{\pi}{2} = \text{Zero}$$

38. Answer (3)

Hint: Power (P) transmitted in the beam = $NE = Nh\nu$

$$\text{Sol.: } P = 2 \times 10^{16} \times 6.63 \times 10^{-34} \times 5 \times 10^{14} \\ = 6.63 \times 10^{-3} \text{ W} = 6.63 \text{ mW}$$

39. Answer (3)

Hint & Sol.: When light travels from rarer to a denser medium, the speed decreases and frequency remain same.

40. Answer (4)

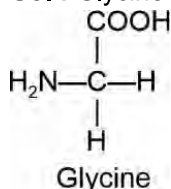
Hint & Sol.: In interference and diffraction, light energy is redistributed. If it reduces in one region, producing a dark fringe, it increases in another region, producing a bright fringe. There is no gain or loss of energy, which is consistent with the principle of conservation of energy.

41. Answer (4)

46. Answer (3)

Hint : Optically active compounds must have chiral carbon or molecular asymmetry.

Sol.: Glycine does not contain chiral centre.



47. Answer (1)

Hint & Sol.: Energy is released when BE/nucleon is increased.

42. Answer (1)

Hint: For n-type, electrons are majority charge carriers.

Sol.: In an extrinsic semiconductor doped with pentavalent impurity, electrons become the majority carriers and holes the minority carriers.

$$n_e \gg n_h$$

43. Answer (3)

Hint: If virtual and erect image is formed so m is +ve $\Rightarrow m = +3$

$$\text{Sol.: } \frac{f}{f-u} = 3$$

$$u = \frac{-20}{3} \text{ cm}$$

44. Answer (2)

Hint: At the instant of closest approach, α -particles is momentarily at rest.

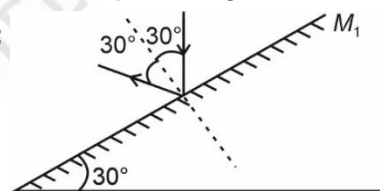
$$\text{Sol.: } \therefore \frac{1}{4\pi\epsilon_0} \frac{(Ze)(2e)}{r_0} = \frac{1}{2}mv^2$$

$$\Rightarrow r_0 = \frac{1}{\pi\epsilon_0} \times \frac{Ze^2}{mv^2}, \Rightarrow r_0 \propto \frac{1}{m}$$

45. Answer (4)

Hint: According to law of reflection, angle of incidence is equal to angle of reflection.

Sol.:



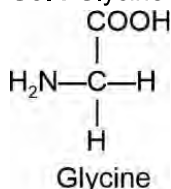
Angle of reflection = 30°

[CHEMISTRY]

46. Answer (3)

Hint : Optically active compounds must have chiral carbon or molecular asymmetry.

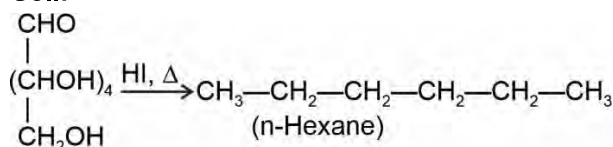
Sol.: Glycine does not contain chiral centre.



47. Answer (1)

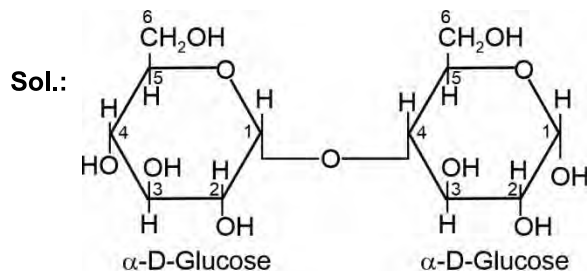
Hint: Glucose on prolong heating with HI, it forms n-hexane.

Sol.:



48. Answer (2)

Hint: Maltose is composed of two α -D-glucose units.



Maltose

49. Answer (3)

Hint: In disaccharides, if the reducing groups of monosaccharides i.e., aldehydic or Ketonic groups are present in the form of acetal then they will not give positive Tollens' test and are called non-reducing sugar.

Sol.: In sucrose, the reducing groups of glucose and fructose are involved in glycosidic bond formation hence will not reduce Tollens' reagent.

50. Answer (2)

Hint & Sol.: Iodine forms intense blue complex with starch.

51. Answer (2)

Hint: On heating metal carbonates in a charcoal cavity, the carbonates are converted to metal oxide, which are coloured residues.

Sol.: As^{3+} gives white residue with the odour of garlic.

52. Answer (4)

Hint: Methyl orange is a weak base and is yellow in colour in the unionised form.

Sol.: Methyl orange is orange red in acidic medium and turns yellow in basic medium.

53. Answer (4)

Hint: Group reagents are used for the separation of cation due to their selective precipitation.

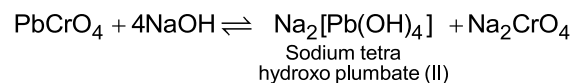
Sol.:

Group	Group reagent
Group-I	Dilute HCl
Group-II	H_2S gas in presence of dil. HCl
Group-III	NH_4OH in presence of NH_4Cl
Group-IV	H_2S in presence of NH_4OH

54. Answer (4)

Hint: PbCrO_4 is insoluble in water and it forms a yellow precipitate.

Sol.:



55. Answer (1)

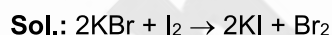
Hint: s-block elements impart characteristic colour to Bunsen flame when heated on platinum loop.

Sol.:

Element		Flame colour (naked eye)
Copper	—	Green with blue centre
Barium	—	Apple green
Strontium	—	Crimson red
Calcium	—	Brick red

56. Answer (2)

Hint: In halogens, higher member can displace lower member from their salt solution.



Reaction is not possible because $\text{Br}^\ominus$ ion is not oxidised in Br_2 with I_2 due to higher electrode potential of I_2 than bromine.

57. Answer (3)

Hint: Percentage purity

$$= \frac{\text{Mass of pure chemical}}{\text{Mass of sample}} \times 100$$

Sol.: 20 ml of 0.1 N KMnO_4 correspond to

= 0.002 eq. KMnO_4 . Thus, 0.002 eq. ferrous ammonium sulphate is reacted. The eq. wt of ferrous ammonium sulphate is 392. 0.002 eq. of ferrous ammonium sulphate corresponds to

$$= 0.002 \times 392 = 0.784 \text{ g}$$

% purity of ferrous ammonium sulphate

$$= \frac{0.784}{1.96} \times 100 = 40\%$$

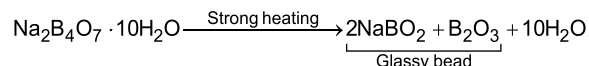
58. Answer (3)

Hint: Molecular orbital diagram of S_2 is similar to that of O_2 .

Sol.: S_2 is paramagnetic with two unpaired electrons in π^* orbitals.

59. Answer (3)

Hint & Sol.:



60. Answer (2)

Hint & Sol.: Colloidal particle ranges from 1 nm to 1000 nm.

61. Answer (2)

Hint & Sol.: $\text{CuSO}_4 + \text{H}_2\text{S} \rightarrow \text{CuS} + \text{H}_2\text{SO}_4$
Black

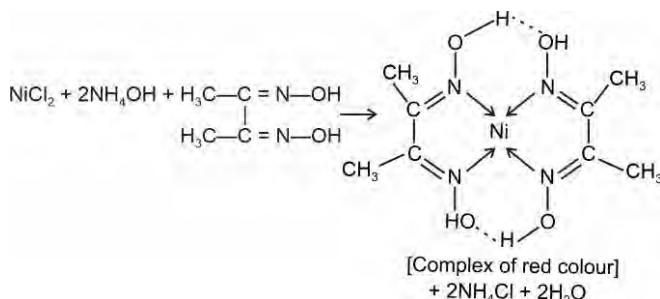
$\text{CuS} + 2\text{HNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + \text{H}_2\text{S}$
Blue solution

$\text{Cu}(\text{NO}_3)_2 + 4\text{NH}_3 \rightarrow [\text{Cu}(\text{NH}_3)_4]^{2+} + 2\text{NO}_3^-$
Deep blue solution

62. Answer (2)

Hint: When dimethyl glyoxime is added to the aqueous solution of nickel chloride and is made alkaline by adding NH_4OH solution, a brilliant red precipitate is obtained.

Sol.:



63. Answer (2)

Hint: Carbon dioxide is a colourless and odourless gas, which turns lime water milky.

Sol.: $\text{Na}_2\text{CO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O} + \text{CO}_2$

64. Answer (1)

Hint: $\text{NH}_4^+ + \text{NaOH} \longrightarrow \text{NH}_3 + \text{H}_2\text{O}$
(Gas X)

Sol.: $\text{NH}_3 + \text{HCl} \longrightarrow \text{NH}_4\text{Cl}$
dense white fumes

$2\text{K}_2\text{HgI}_4 + \text{NH}_3 + 3\text{KOH} \Rightarrow \text{HgO} \cdot \text{Hg}(\text{NH}_2)_2 + 7\text{KI} + 2\text{H}_2\text{O}$
(Brown-precipitate)

65. Answer (4)

Hint: H_2S in presence of NH_4OH is the group reagent for group IV.

Sol.: When H_2S is passed from Zn salt solution in presence of NH_4OH , ZnS is formed, it is a white ppt.

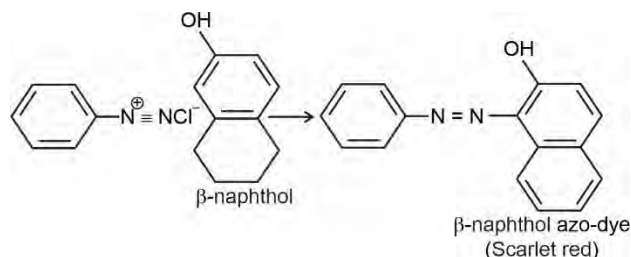
66. Answer (4)

Hint & Sol.: AgI is yellow and insoluble in NH_3 solutions.

67. Answer (4)

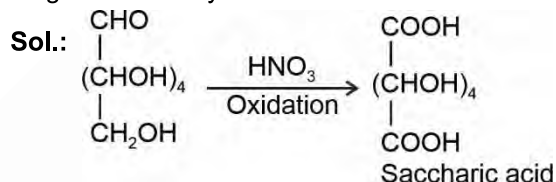
Hint: On adding of β -naphthol to diazonium salt, coupling reaction takes place to form a scarlet coloured dye which is sparingly soluble in water.

Sol.:



68. Answer (2)

Hint: On oxidation with nitric acid, glucose as well as gluconic acid yield saccharic acid.



69. Answer (3)

Hint & Sol.: Vitamin B1 is known as thiamine

Vitamin B2 is known as Riboflavin

Vitamin B6 is known as Pyridoxine

Vitamin C is known as Ascorbic acid

70. Answer (2)

Hint: When the polypeptide chains run parallel and are held together by hydrogen and disulphide bonds, then fibre-like structure is formed.

Sol.: Keratin and myosin are fibrous proteins.

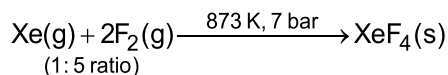
71. Answer (1)

Hint & Sol.:

Formula	Physical state and colour
ClF_3	Colourless gas
BrF_3	Yellow green liquid
IF_3	Yellow powder
ICl_3	Orange solid

72. Answer (2)

Hint & Sol.: Xenon forms binary fluorides by the direct reaction of elements under appropriate experimental conditions.



73. Answer (4)

Hint & Sol.:

Compound Name	Formula
Tear gas	CCl_3NO_2
Phosgene	COCl_2
Laughing gas	N_2O
Mustard gas	$\text{ClCH}_2\text{CH}_2\text{SCH}_2\text{CH}_2\text{Cl}$

74. Answer (4)

Hint: $P_4 + 3NaOH + 3H_2O \rightarrow PH_3 + 3NaH_2PO_2$
(sodium hypophosphite)

Sol.: • It gives phosphonium compounds with acids
 $PH_3 + HBr \rightarrow PH_4Br$

• It is a colourless gas with rotten fish smell

75. Answer (3)

Hint: Hot concentrated sulphuric acid is a moderately strong oxidising agent.

Sol.:

$$\begin{array}{ccc} & \text{+6} & \text{+4} \\ & \text{-----} & \\ Cu + 2H_2SO_4 (\text{conc.}) & \rightarrow & CuSO_4 + SO_2 + 2H_2O \end{array}$$

76. Answer (2)

Hint: On moving down the group bond length increases $B.E. \propto \frac{1}{\text{Bond length}}$

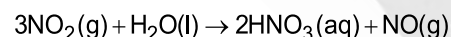
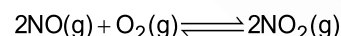
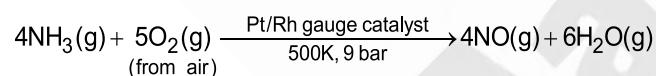
Sol.:

Hydrides of group 15	Bond dissociation energy (kJ/mol)
NH ₃	389
PH ₃	322
AsH ₃	297
SbH ₃	255

77. Answer (4)

Hint: In Ostwald's process, NO formed combines with O₂ to give NO₂ which is dissolve in water to give HNO₃.

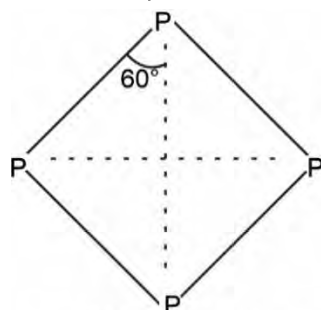
Sol.: This process is based on catalytic oxidation of NH₃.



78. Answer (2)

Hint: Red phosphorus is insoluble in water as well as in carbon disulphide.

Sol.:



Red phosphorus is obtained by heating white phosphorus at 573 K in an inert atmosphere for several days.

79. Answer (4)

Hint: Borax is Na₂B₄O₇·10H₂O

Sol.: In borax two boron atoms are sp² hybridised while other two are sp³ hybridised.

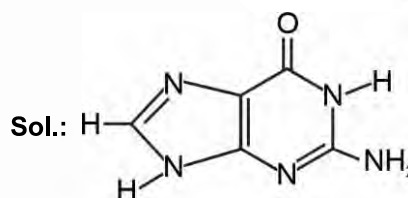
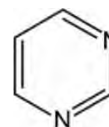
80. Answer (4)

Hint: Compound having vacant orbital act as Lewis's acid.

Sol.: NH₃ act as Lewis base due to presence of lone pair.

81. Answer (3)

Hint: Structure of pyrimidine is



(Guanine); It is purine type base.

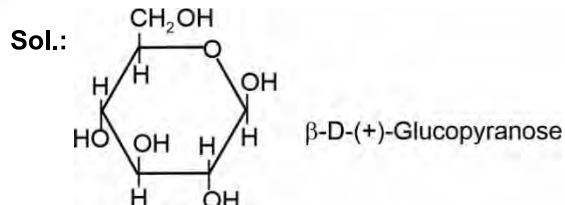
82. Answer (1)

Hint: In amylopectin, branching occurs by C1-C6 glycosidic linkage.

Sol.: Amylose is water soluble components which constitutes about 15-20% of starch. Amylopectin is insoluble in water and constitutes about 80-85% of starch. It is a branched chain polymer of α-D-glucose units in which chain is formed by C1-C4 glycosidic linkage whereas branching occurs by C1-C6 glycosidic linkage.

83. Answer (1)

Hint: β-D-(+)-Glucopyranose contains five chiral carbon atoms.



84. Answer (3)

Hint: Vitamin A, D, E and K are fat soluble.

Sol.: B group vitamins and vitamin C are water soluble.

85. Answer (4)

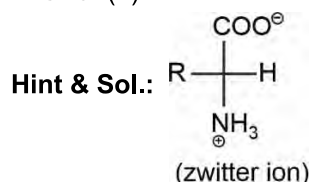
Hint: Complete hydrolysis of DNA yields a pentose sugar, phosphoric acid and nitrogen containing heterocyclic compound.

Sol.: Sugar moiety present in RNA molecule is β-D-ribose.

86. Answer (2)

Hint & Sol.: During denaturation of proteins, secondary and tertiary structure are destroyed but primary structure remains intact.

87. Answer (1)



88. Answer (2)

Hint: The amino acids, which can be synthesised in the body, are known as non-essential amino acid.

Sol.: Tyrosine is a non-essential amino acid.

89. Answer (1)

Hint: Glucose does not form the hydrogen sulphite addition product with NaHSO_3

Sol.: Cellulose is a straight chain polysaccharide composed only of β -D-glucose units.

90. Answer (4)

Hint: Lactose is a disaccharide which is found in milk.

Sol.: Lactose on hydrolysis gives β -D-galactose and β -D-glucose. It is a reducing sugar.

[BOTANY]

91. Answer (2)

Hint: Sarguja, Chanda and Bastar are the places of sacred groves belonged to Madhya Pradesh.

Sol.: • Khasi and Jaintia Hills are in Meghalaya.

• Aravalli Hills are in Rajasthan.

• Western Ghat regions are in Karnataka and Maharashtra.

92. Answer (4)

Hint: Anthropogenic ecosystems are created and maintained by human being and they have little diversity.

Sol.: Anthropogenic ecosystem possesses high productivity with simple food chain.

93. Answer (3)

Hint: Key industry animals are first order consumers.

Sol.: Producers are also known as transducers because they convert solar energy into the chemical energy.

94. Answer (3)

Hint: Some of the key functional aspects of the ecosystem are productivity and decomposition.

Sol.: Stratification is the structural feature of the ecosystem.

95. Answer (4)

Hint: Primary productivity is the rate of biomass production by plants per unit area over a time period.

Sol.: Vertical stratification is more diverse in forests than in desert ecosystem as forests have more types of vegetation and form more different layers as compared to desert.

96. Answer (1)

Hint: Anthropogenic ecosystem is maintained by human beings.

Sol.: Estuaries (salt water) is a natural, aquatic ecosystem. Crop field, garden and aquarium are created by human beings.

97. Answer (4)

Hint: Species of birds (1300) are more in numbers as compared to mammals (427) in the Amazonian rain forests.

Sol.:

Taxa	Number of species
Mammals	427
Birds	1300
Reptiles	378
Amphibians	427
Fishes	3000

98. Answer (3)

Sol.: Primary productivity is expressed in terms of weight as $(\text{gm m}^{-2})\text{yr}^{-1}$ or energy as $(\text{kcal m}^{-2})\text{yr}^{-1}$.

99. Answer (3)

Hint: Temperature plays an important role in both latitudinal and altitudinal distribution of organisms.

Sol.: Mango, a tropical and subtropical tree plant is not found in temperate countries like Canada and Germany.

100. Answer (4)

Hint: Euryhaline are organisms which tolerate a wide range of salinities.

Sol.: Stenohaline are those organisms, which are restricted to a narrow range of salinities.

101. Answer (1)

Hint: Warm-blooded animals (birds and mammals) have larger body size in cold climates than in hotter areas. This is called Bergmann's rule.

Sol.: Mammals from colder climates generally have shorter ears and limbs to minimise heat loss. This is called as Allen's rule.

- **Rensch's rule:** In colder climate, birds possess narrow and acuminate wings as compared to broader wings of birds found in warmer areas.

- **Jordan's rule:** As the temperature is lowered, some fishes possess larger size with larger number of vertebrae.

102. Answer (1)

Hint: Emigration is the number of individuals of the population who had left the habitat and gone elsewhere during the time period under consideration.

Sol.: Natality refers to the number of births during a given period in the population that are added to initial density.

- Mortality is the number of deaths in the population during given period.

- Immigration is the number of individuals of the same species that have come into the habitat from elsewhere during time period under consideration.

103. Answer (4)

Hint: Sex ratio is an important population attribute.

Sol.: Birth rate and death rates are important population attributes.

An individual may have birth and death.

104. Answer (4)

Hint: In predation, only one species (predator) benefits and the interaction is detrimental to the other species (Prey).

Sol.: In competition, both the interacting species are negatively affected.

105. Answer (1)

Hint: Mineralisation is the release of inorganic substances from organic matter.

Sol.: Standing crop is the amount of all the living material present in an ecosystem per unit area at a given time.

- Standing state or quality is the amount of all the inorganic substances present in an ecosystem per unit area at a given time.

106. Answer (3)

Hint: Ecosystem diversity is high in India, because of the occurrence of large number of ecosystems like desert, rain forests and many rivers etc.

Sol.: Species diversity is the product of species richness and species evenness.

107. Answer (3)

Hint: An urn-shaped age pyramid shows declining population.

Sol.: Urn-shaped age pyramid shows higher number of reproductive individuals as compared to pre-reproductive individuals.

108. Answer (3)

Hint: In species-area relationship equation, species richness (S) is directly proportional to the area (A).

Sol.: The correct equation of species-area relationship on a logarithmic scale is

$$\log S = \log C + Z \log A$$

109. Answer (4)

Hint: The hotspots are richest and most threatened reservoirs of plants and animals life on earth.

Sol.: Collection of forest products, harvesting of timber and tilling of land are allowed in sanctuaries. Drugs and industrial products are included in narrowly utilitarian categories while aesthetic pleasure is included in broadly utilitarian categories.

110. Answer (4)

Hint: Humans have always been dependent on nature for food and shelter, but when need turns to greed it leads to overexploitation of natural resources.

Sol.: Habitat loss and fragmentation is the most important cause driving animals and plants to extinction.

111. Answer (3)

Hint: Communities with more species tend to be more stable.

Sol.: A stable community should not show too much variation in productivity from year to year.

112. Answer (3)

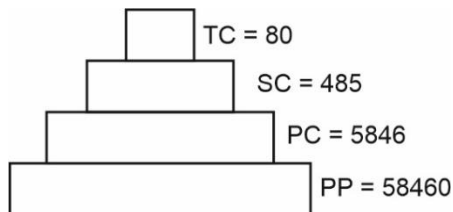
Hint: Predators help in maintaining species diversity in a community.

Sol.: Biological pest control methods used in agriculture are based on the ability of predator to regulate prey population.

113. Answer (3)

Hint: Pyramid of number in aquatic ecosystem can be upright.

Sol.:



114. Answer (4)

Hint: Decomposers are also called reducers as they are capable of degrading the dead organism.

Sol.: Decomposers secrete digestive enzymes which convert complex organic substances into simpler ones.

115. Answer (2)

Hint: Any species growing exponentially under unlimited resource conditions can reach enormous population densities in a short time.

Sol.: The integral form of exponential growth equation will be

$$N_t = N_0 e^{rt}$$

N_t = population density after 't' time

N_0 = population density at time zero

e = The base of natural logarithms.

116. Answer (4)

Sol.: By these population characters, the density of population (N) at time 't' can be expressed after a period of time 't + 1'

$$\text{As } N_{t+1} = N_t + [(B + I) - (D + E)]$$

$$B = 350$$

$$I = 20$$

$$D = 340$$

$$E = 30$$

$$N_{t+1} = N_t + [(350 + 20) - (340 + 30)]$$

$$N_{t+1} = \text{Zero}$$

117. Answer (3)

Hint: Archaeobacteria sustain their life under extremes of temperature due to branched chain lipid in membrane.

Sol.: Bear escape in time during winter by a process called hibernation.

118. Answer (1)

Hint: Over-exploitation of particular species reduces size of its population to an extent so that it becomes vulnerable to extinction.

Sol.: New species entering a geographical region are called exotic or alien or non-native species e.g., Nile perch, was introduced into Lake Victoria of East Africa.

119. Answer (4)

Hint: Energy flow in an ecosystem is unidirectional.

Sol.: Energy flow follows laws of thermodynamics.

120. Answer (4)

Sol.: Sparrow occupies more than one trophic level in the same ecosystem at the same time.

121. Answer (1)

Hint: The effect of reduction in biodiversity has been explained by Paul Ehrlich through Rivet popper hypothesis.

Sol.: • Species-area relationships was given by Alexander von Humboldt.

• Term biodiversity was popularised by Edward Wilson.

• According to Robert May, the number of global species diversity is about 7 million.

122. Answer (3)

Hint: Rabbit is herbivore and it is primary consumer in grassland ecosystem.

Sol.: Grass → Grasshopper → Frog → Snake → Eagle.

123. Answer (1)

Hint: Rate of decomposition of detritus is controlled by chemical nature of detritus and number of climatic factors.

Sol.: Temperature and soil moisture are the most important climatic factors that regulate decomposition through their effect on the activities of soil microbes.

An optimum moisture helps in quicker decomposition of detritus.

At a temperature of more than 25°C, decomposers are very active in soils having good moisture and aeration.

124. Answer (2)

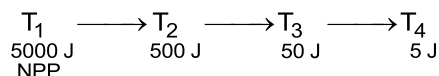
Hint: A biosphere reserve consists of core, buffer and transition zones.

Sol.: The buffer zone surrounds the core area and is managed to accommodate a greater research and educational activities.

125. Answer (3)

Hint: The transfer of energy from one trophic level to another trophic level is accompanied by loss of energy at each level or step.

Sol.: Only 10% energy is transferred from one trophic level to another trophic level.



So, the amount of energy which gets stored and fixed in tertiary consumer will maximally be 5 J.

126. Answer (3)

Hint: Very less amount of energy is present at highest trophic level.

Sol.: Pyramid of energy is always upright in all ecosystem. Because only 10% energy is transferred from one trophic level to another trophic level and remaining 90% is consumed in respiration and other metabolic activities.

127. Answer (2)

Hint: Mineralisation is the release of inorganic substances from organic matter.

Sol.: Humification is the process of decomposition of detritus to form humus. Humus is a dark coloured, amorphous, more or less decomposed organic matter rich in cellulose, lignin etc.

In leaching, part of water-soluble substances present in the fragmented and decomposing detritus go down into the soil horizon by percolating water and get precipitated as unavailable salts.

128. Answer (3)

Hint: Ecological pyramids assume a simple food chain and does not accommodate a food web.

Sol.: Saprophytes, decomposers, microbes and detritivores are not given any place in ecological pyramids.

129. Answer (3)

Hint: Mycorrhiza is a mutualistic relationship between fungi and roots of higher plants.

Sol.: Commensalism – orchids growing on mango tree.

Parasitism – Ticks on dogs.

130. Answer (2)

Hint: Stratification is the vertical distribution of different species occupying different levels.

Sol.: Species composition and stratification are the structural components of an ecosystem.

131. Answer (4)

Hint: In accordance with life styles, parasites have evolved a number of special adaptations.

Sol.: Parasites show high reproductive capacity, loss of digestive system, etc. as adaptations to survive in host.

132. Answer (3)

Hint: Bear escapes in time during winter by a process caused hibernation.

Sol.: A stage of suspended development found in many zooplankton species is called diapause.

133. Answer (1)

Hint: Presence of thorns are protective mechanisms shown by plants.

Sol.: Plants can avoid their predators like phytophagous insects because plants have evolved an astonishing variety of morphological and chemical defence against herbivores.

134. Answer (4)

Hint: Dodo was extinct from Mauritius.

Sol.: Steller's sea cow was the most recent extinction from Russia.

135. Answer (2)

Sol.: The World Summit on Sustainable Development was held in 2002 in Johannesburg, South Africa.

[ZOOLOGY]

136. Answer (1)

Hint: Bacterial transformation

Sol.: In order to force bacteria to take up the plasmid, the bacterial cells are made competent by treating them with the specific concentration of a divalent cation like Ca^{2+} . DNA is a negatively charged molecule.

137. Answer (2)

Hint: Part of chemical engineering

Sol.: Two techniques enabled the birth of modern biotechnology.

a. Genetic engineering → Techniques that alter the chemistry of genetic material (DNA and RNA) to introduce these into host organisms and change the phenotype of plant / animal.

b. Bioprocess engineering → Maintenance of the sterile environment for mass production of biotechnological products.

138. Answer (2)

Hint: Use of only natural resources

Sol.: The applications of biotechnology include the early diagnosis, genetically modified crops for

agriculture, processed food, bioremediation, waste treatment.

Traditional hybridisation procedures used in plant and animal breeding, very often lead to inclusion and multiplication of undesirable genes along with the desired genes.

139. Answer (1)

Hint: Features that are required to facilitate cloning

Sol.: *ori* sequence is responsible for controlling the copy number of the linked DNA.

Normal *E.coli* cells do not carry resistance against ampicillin and tetracycline antibiotics. Vectors need to have very few, preferably single restriction site for commonly used restriction enzymes. Presence of more than one recognition sites within the vector will generate several fragments which will complicate the gene cloning.

140. Answer (2)

Hint: Processes of rDNA

Sol.: Recombinant DNA technology involves several steps in specific sequence such as isolation of DNA, fragmentation of DNA by endonuclease, ligation of the DNA fragment into a vector, transferring the recombinant DNA into the host, culturing the host cells in a medium at large scale and extraction of the desired product.

141. Answer (4)

Hint: Exclude the nematode

Sol.: Pomato is not a GMO but formed by somatic hybridisation but Rosie is a GMO.

Meloidogyne – Nematode

Bacillus thuringiensis – Microbe

142. Answer (1)

Hint: Tissue culture

Sol.: If the plant is infected with a virus, the meristem (apical and axillary) is free of virus. Hence, one can remove the meristem and grow it *in vitro* to obtain virus free plants.

143. Answer (3)

Hint: Can be used for treating night blindness

Sol.: Genetic modification in plants are useful as they

- (i) Made crops more tolerant to abiotic stresses
- (ii) Reduced reliance on chemical pesticides.
- (iii) Enhanced nutritional value of food, e.g. Golden rice, *i.e.* Vitamin 'A' enriched rice.

144. Answer (3)

Hint: Gene gun

Sol.: In plants, cells are bombarded with high velocity microparticles of gold or tungsten coated with DNA in a method known as biolistics or gene gun. Microinjection is used for animals such as *Rattus*, nematodes and *Meloidogyne*.

145. Answer (2)

Hint: Reporter enzyme

Sol.: Selection of recombinants due to inactivation of antibiotics is a cumbersome procedure. Therefore, alternative selectable markers have been developed which differentiate recombinants from non-recombinants on the basis of their ability to produce colour in the presence of a chromogenic substrate. A recombinant DNA is inserted within the coding sequence of an enzyme, β -galactosidase.

Reporter enzyme such as β -galactosidase can be used to monitor transformation of host cells.

146. Answer (1)

Hint: Facilitate the mixing of the reactor contents.

Sol.: Stirred tank bioreactor contains flat bladed impeller, curved base with foam breaker to facilitate the mixing of the reactor contents.

When protein encoding gene is expressed in heterologous host, protein produced is called recombinant protein.

First recombinant DNA was constructed by linking a gene encoding the antibiotic resistance with native plasmid of *Salmonella typhimurium*.

147. Answer (3)

Hint: Used for separation and isolation of DNA fragments.

Sol.: DNA fragments are negatively charged and can be separated by forcing them to move towards the anode under an electric field through a medium.

DNA fragments separate according to their size under sieving effect by agarose gel. Separate bands of DNA are cut out and extracted from the gel piece by elution.

148. Answer (1)

Hint: Choose the order to which cotton bollworm belongs to

Sol.:

Lepidopterans	–	Tobacco budworm, armyworm
Coleopterans	–	Beetles
Dipterans	–	Flies, mosquitoes

149. Answer (2)

Hint: RNA interference

Sol.: RNAi involves silencing of a specific mRNA due to complementary dsRNA molecule that binds to and prevents translation of the mRNA (silencing).

150. Answer (1)

Hint: Insulin consists of peptides linked together by disulfide bridges.

Sol.:

GM plants	–	Fuels and pharmaceuticals
ADA deficiency	–	Genetically engineered lymphocytes
Mature insulin	–	Two interchain disulphide bonds present in between chain A and B and one intrachain disulphide bond present in chain A

151. Answer (3)

Hint: Insecticidal protein

Sol.: Some strains of *Bacillus thuringiensis* produce proteins that kill certain insects such as lepidopterans (Tobacco budworm, Armyworm), coleopterans (Beetles) and dipterans (Flies, Mosquitoes).

Agrobacterium is a pathogen which causes crown gall disease in dicots.

Meloidogyne incognita is a nematode which infects the roots of tobacco plants.

Disarmed retrovirus can be used as vector to deliver foreign DNA in an animal cell.

152. Answer (2)

Hint: Exclude the treatment after birth

Sol.: Children suffering from ADA deficiency can be cured by bone marrow transplantation, enzyme replacement therapy, gene therapy but problem with all these methods is that they are not completely curative in nature.

However, if the gene isolated from marrow cells producing ADA is introduced into cells at early embryonic stages then it could be a permanent cure.

153. Answer (4)

Hint: *Agrobacterium tumefaciens* possesses Ti plasmid

Sol.: *Agrobacterium tumefaciens*, a pathogen of several dicot plants is able to deliver a piece of DNA known as 'T-DNA' to transform normal plant cells into a tumor.

Ti plasmid (Tumor inducing plasmid of *Agrobacterium*) has been modified into a cloning vector which is no more pathogenic to the plants.

Meloidogyne is a nematode which infects tobacco plants.

154. Answer (2)

Hint: Safety testing of polio vaccine

Sol.: Transgenic mice are being developed for use in testing the safety of vaccines before they are used on humans. Transgenic mice are being used to test the safety of the polio vaccine. If successful and found to be reliable, they could replace the use of monkeys to test the safety of batches of vaccine.

155. Answer (2)

Hint: Steps in autoradiography

Sol.: A single stranded DNA or RNA, tagged with a radioactive molecule (probe) is allowed to hybridised to its complementary DNA in a clone of cells followed by detection using radiography. The clone having the mutated gene will hence not appear on the photographic film, because the probe will not have complementarity with the mutated gene.

156. Answer (2)

Hint: Treatment of emphysema

Sol.: Insulin is given for treatment of diabetes. α -1-antitrypsin is given to treat emphysema.

α -lactalbumin is present in the milk produced by first transgenic cow, Rosie.

157. Answer (2)

Hint: Belongs to the order Lepidoptera**Sol.:** Bt toxins are insect-group specific.

Proteins encoded by the genes *cryI*Ac and *cryII* Ab control the cotton bollworms and that of *cry I*Ab controls corn borer.

158. Answer (4)

Hint: Formation of sticky ends

Sol.: Restriction enzymes cut the strand of DNA little away from centre of the palindromic sequence and leave sticky ends on each strand. The stickiness of ends facilitates the action of enzyme DNA ligase when both alien DNA and plasmid are cut by same restriction enzyme.

159. Answer (2)

Hint: Restriction enzymes are nucleases

Sol.: Nucleases are of two types : exonucleases and endonucleases. Exonucleases remove nucleotides from the ends of the DNA whereas endonucleases make cuts at specific positions within the DNA.

Restriction endonucleases are naturally present in prokaryotic cells.

160. Answer (3)

Hint: Methylase and restriction endonuclease

Sol.: In 1963, two enzymes were discovered for restricting the growing of bacteriophage in *E.coli*. One (methylase) added methyl groups to DNA of bacteria and the other (endonuclease) cuts the DNA of bacteriophage.

161. Answer (2)

Hint: Exclude the events in 1990s

Sol.: In the year 1963, the two enzymes responsible for restricting the growth of bacteriophage in *Escherichia coli* were isolated. One of those enzyme was restriction endonuclease. The first restriction endonuclease *HindIII* was isolated and characterised five years later.

In 1990, first clinical gene therapy was given to a 4-years old girl.

In 1997, first transgenic cow, Rosie was produced.

162. Answer (1)

Hint: Continuous culture system

Sol.: The initial phase is the lag phase where bacteria are metabolically active but not dividing. The exponential or log phase is the time of exponential growth. To obtain large culture, cells are kept in log phase.

In the stationary phase, growth reaches a plateau.

In death or decline phase, growth of bacteria decreases again.

163. Answer (4)

Hint: More than the normal human body temperature

Sol.: Heat shock for bacterial transformation is done at 42°C.

Denaturation of nucleic acids in PCR is done at 94°C.

Thermophilic enzymes remain stabilised upto 80-90°C and normal human body temperature stays in the range of 36-37°C.

164. Answer (1)

Hint: Spooling

Sol.: DNA isolation can be achieved by treating the bacterial cells / plant or animal tissue with enzymes such as lysozyme (bacterial), cellulase (plant) or chitinase (fungus). DNA ligase assists in ligation of fragments of DNA.

Ribonuclease can be used to remove RNA. Purified DNA ultimately precipitates out after the addition of chilled ethanol and can be seen as collection of fine threads in the suspension.

165. Answer (4)

Hint: Used as a selectable marker.

Sol.:

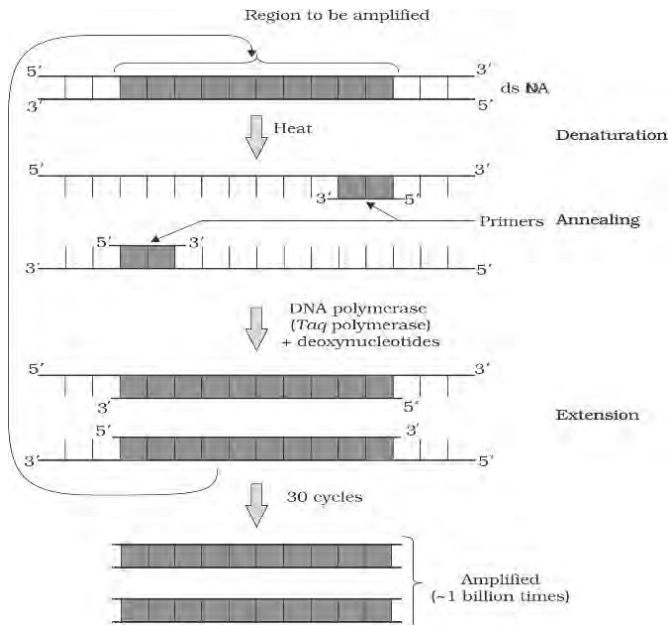
Transposons	–	Mobile genetic elements
Agarose	–	Natural polymer extracted from sea weeds, used as matrix in gel electrophoresis
Lysozyme	–	Enzyme used to breakdown bacterial cell wall
Chloramphenicol	–	Antibiotic

166. Answer (2)

Hint: Insertional inactivation

Sol.: β -galactosidase produce blue colour in presence of chromogenic substrate. Now, if foreign DNA is inserted within β -galactosidase coding sequence, it will lead to insertional inactivation due to which recombinants will produce white coloured colonies and non-recombinants will produce blue coloured colonies in presence of chromogenic substrate.

167. Answer (3)

Hint: Guanylic acid is a nucleotide.**Sol.:**

Guanosine is a nucleoside and cytosine is a nitrogenous base.

168. Answer (3)

Hint: Used for animal cells

Sol.: In vectorless method known as micro-injection, recombinant DNA is directly injected into the nucleus of an animal cell.

169. Answer (4)

Hint: Mature insulin has one intrachain disulfide bond

Sol.: Insulin used for diabetes earlier was extracted from pancreas of slaughtered cattle and pigs. Insulin is synthesised as a pro-hormone and before it becomes fully mature, an extra stretch called C peptide is removed and mature form contains two short polypeptides called A and B.

In 1983, Eli Lilly, an American company, prepared two DNA sequences corresponding to A and B chains of human insulin and introduced them in plasmids of *E. coli* to produce human insulin.

170. Answer (3)

Hint: Exclude the Bt crop.

Sol.: In 1997, American company got patent over Basmati rice and same attempts have been made to patent a product and process based on Indian traditional herbal medicine such as neem and turmeric.

171. Answer (1)

Hint: Use of agrochemicals.

Sol.: The Green Revolution succeeded in tripling the food supply but yet it was not enough to feed the growing human population. Increased yields have partly been due to the use of improved crop varieties, but mainly due to the use of better management practices and use of agrochemicals (fertilisers and pesticides).

172. Answer (4)

Hint: Conventional method of diagnosis

Sol.: Recombinant DNA technology, Polymerase Chain Reaction and Enzyme Linked Immunosorbent Assay are some of the techniques that serve the purpose of early diagnosis.

Urine analysis and serum analysis are conventional method of diagnosis.

173. Answer (3)

Hint: Include the genetic disorder

Sol.: Today, transgenic models exist for human diseases such as cancer, Alzheimer's, cystic fibrosis and Rheumatoid arthritis.

174. Answer (2)

Hint: Insertional inactivation

Sol.: The ligation of foreign DNA at *EcoRI* site will not lead to any insertional inactivation of the antibiotic resistance gene. Due to which recombinant *E. coli* will be resistant against both tetracycline and ampicillin antibiotics.

175. Answer (1)

Hint: Series of processes that also include marketing of the product

Sol.: Transformation is a procedure through which a piece of DNA is introduced in a host cell.

Gel electrophoresis → Technique utilised for separation of DNA fragments after digestion by endonucleases.

Upstream processing → Includes all the processes upto formation of recombinant protein.

176. Answer (3)

Hint: Endonucleases are found in prokaryotes.

Sol.: First letter of the name of restriction enzyme for e.g., *BamHI* comes from the genus which is *Bacillus*, second two letters comes from species which is *amyloliquefaciens*.

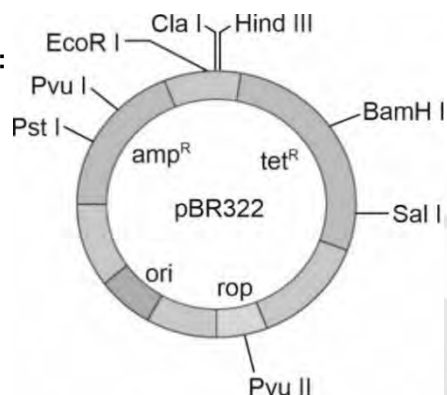
The letter 'H' is derived from the strain. Roman numbers following the name which is 'I' indicates the order in which enzyme was isolated from that strain of bacteria.

177. Answer (3)

Hint: DNA ligase**Sol.:** DNA ligases also known as molecular glue belong to class VI of enzymes, catalyse the linking together of alien DNA and cloning vector.

Restriction endonucleases also known as molecular scissors help in cutting the DNA at specific locations.

178. Answer (3)

Hint: Exclude the enzymes whose restriction site is within tet^R **Sol.:**

179. Answer (2)

Hint: Upkeeps the normal standards of GM research**Sol.:** Genetic Engineering Approval Committee has been setup by Indian government to make decisions regarding the validity of GM research and safety of introducing GM-organisms or products for public services.

NACO → National AIDS Control Organisation

EFB → European Federation of Biotechnology

WHO → World Health Organisation.

180. Answer (2)

Hint: Also constructed artificial recombinant DNA for the first time.**Sol.:** Stanley Cohen developed a method of removing plasmids from the cell and then reinserting them in other cells.Herbert Boyer observed about restriction enzymes of the *E.coli* and their useful properties.

S.L. Miller did a simulation experiment for chemosynthesis.

Watson and Crick gave double helical model of B-DNA.

□ □ □